Southern Nuclear

Regulatory Affairs

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Docket Nos. 52-025
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ATTN: Document Control Desk
U. S. Nuclear Regulatory Commission
Washington, D. C. 20555-0001

Southern Nuclear Operating Company
Vogtle Electric Generating Plant – Units 3 and 4
License Amendment Request: Change to Technical Specification 3.5.2,
"Core Makeup Tanks (CMTs) – Operating" Boron Concentration Limits

Pursuant to 10 CFR 52.98(c) and in accordance with 10 CFR 50.90, Southern Nuclear Operating Company (SNC) hereby requests a license amendment to the Technical Specifications (TS) for Vogtle Electric Generating Plant (VEGP) Units 3 and 4, Combined Licenses (COLs) NPF-91 and NPF-92, respectively. This license amendment request (LAR) proposes changes to TS 3.5.2, Core Makeup Tanks (CMTs) - Operating, Surveillance Requirement (SR) 3.5.2.4, to make permanent and applicable to both CMTs for both units the minimum allowable CMT boron concentration and the minimum average CMT boron concentration that was previously approved for VEGP Unit 4 [ADAMS Accession No. ML25225A229]. This request also proposes changes to delete the temporary Notes that were added to Limiting Condition for Operation (LCO) and SR 3.5.2.4 by the previously approved amendment for VEGP Unit 4. To support the CMT boron concentration limit change, this request also presents a revision to the analysis of the inadvertent operation of a CMT during power operation event, as discussed in Updated Final Safety Analysis Report (UFSAR) subsection 15.5.1.

These changes were previously discussed with the NRC Staff at a public observation pre-submittal conference call on January 28, 2026 [ML26029A451].

The enclosure to this letter provides the description, technical evaluation, regulatory evaluation (including the Significant Hazards Consideration Determination) and environmental considerations for the proposed changes.

SNC requests approval of the proposed LAR no later than nine months from acceptance. SNC expects to implement the proposed amendment within 60 days of issuance.

This letter contains no regulatory commitments. This letter has been reviewed and determined not to contain security-related information.

In accordance with 10 CFR 50.91, a copy of this application, with enclosures, is being provided to the designated State of Georgia official.

If you have any questions, please contact Ryan Joyce at 205-992-6468.

I declare under penalty of perjury that the foregoing is true and correct.
Executed on the 26th day of February 2026.

Respectfully submitted,

Jamie Coleman Digitally signed by JAMIEMCO
Date: 2026.02.26 08:54:13 -06'00'

Jamie M. Coleman
Director, Regulatory Affairs
Southern Nuclear Operating Company

Enclosure: Evaluation of the Proposed Change

cc: NRC Regional Administrator, Region II
NRR Project Manager – Vogtle 3 & 4
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RType: VND.LI.L00

ENCLOSURE to NL-25-0455
Evaluation of the Proposed Change

Subject: License Amendment Request: Change to Technical Specification 3.5.2, "Core Makeup Tanks (CMTs) – Operating" Boron Concentration Limits

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1. SUMMARY DESCRIPTION

Pursuant to 10 CFR 52.98(c) and in accordance with 10 CFR 50.90, Southern Nuclear Operating Company (SNC) hereby requests a license amendment to the Technical Specifications (TS) for Vogtle Electric Generating Plant (VEGP) Units 3 and 4, Combined Licenses (COLs) NPF-91 and NPF-92, respectively. This license amendment request (LAR) proposes changes to TS 3.5.2, Core Makeup Tanks (CMTs) - Operating, Surveillance Requirement (SR) 3.5.2.4, to make permanent and applicable to both CMTs for both units the minimum allowable CMT boron concentration and the minimum average CMT boron concentration that was previously approved for VEGP Unit 4 [ADAMS Accession No. ML25225A229]. This request also proposes changes to delete the temporary Notes that were added to Limiting Condition for Operation (LCO) 3.5.2 and SR 3.5.2.4 by the previously approved amendment for Unit 4. To support the CMT boron concentration limit change, this request also presents a revision to the analysis of the inadvertent operation of a CMT during power operation event, as discussed in Updated Final Safety Analysis Report (UFSAR) subsection 15.5.1.

2. DETAILED DESCRIPTION

2.1 System Design and Operation

As described in UFSAR Section 6.3, the passive core cooling system (PXS) performs the primary function to provide emergency core cooling following postulated design basis events. The PXS is a seismic Category I, safety-related system and consists of two CMTs, two accumulators, the in-containment refueling water storage tank (IRWST), the passive residual heat removal heat exchanger, pH adjustment baskets, and associated piping, valves, instrumentation, and other related equipment. The automatic depressurization system valves and spargers, which are part of the reactor coolant system (RCS), also provide important passive core cooling functions. The two redundant CMTs contain sufficient borated water to provide RCS reactivity and inventory control for the design basis accidents (DBAs), including both loss of coolant accident (LOCA) events and non-LOCA events.

The CMTs are cylindrical tanks with hemispherical upper and lower heads. They are made of carbon steel and clad on the internal surfaces with stainless steel. They are located in containment at an elevation slightly above the reactor coolant loops. Each tank contains borated water. During normal operation, the CMTs are maintained at RCS pressure through a normally open pressure balance line from the cold leg.

The outlet line from each CMT is connected to one of two direct vessel injection lines, which provides an injection path for the borated water supplied by the CMT. The outlet line from each CMT is isolated by parallel, normally closed, fail open valves. Upon receipt of an actuation signal, these four valves open to align the CMTs to the RCS.

The CMTs will inject to the RCS as inventory is lost and steam or reactor coolant is supplied to the CMT to displace the water that is injected. Steam or reactor coolant is provided to the CMT through the cold leg balance line, depending upon the specific event that has occurred. The inlet line from the cold leg is sized for LOCA events, where the cold legs become voided and higher CMT injection flows are required.

The CMT size and injection capability are selected to provide adequate RCS boration and safety injection for the limiting DBA. One CMT is adequate for this function during a small break LOCA where one CMT completely spills via the pipe break. The core makeup tanks also provide

RCS makeup and boration during events not involving loss of coolant when the normal makeup system is unavailable or insufficient. The boration capability of these tanks provides adequate core shutdown margin following a steam line break.

The pressure balance line from the cold leg is normally open to maintain the CMTs at RCS pressure, which prevents water hammer upon initiation of CMT injection. The cold leg pressure balance line is connected to the top of the cold leg and is routed continuously upward to the high point near the CMT inlet. The normal water temperature in this line will be hotter than the discharge line.

The outlet line from the bottom of each CMT provides an injection path to one of the two direct vessel injection lines, which are connected to the reactor vessel downcomer annulus. Upon receipt of a safeguards actuation signal, the two parallel valves in each discharge line open to align the associated CMT to the RCS.

There are two operating processes for the CMTs, steam-compensated injection and water recirculation. During steam-compensated injection, steam is supplied to the CMTs to displace the water that is injected into the RCS. This steam is provided to the CMTs through the cold leg pressure balance line. The cold leg line only has steam flow if the cold legs are voided.

During water recirculation, hot water from the cold leg enters the CMTs, and the cold water in the tank is discharged to the RCS. This results in RCS boration and a net increase in RCS mass.

The operating process for the CMTs depends on conditions in the RCS, primarily voiding in the cold leg. When the cold leg is full of water, the cold leg pressure balance line remains full of water and the injection occurs via water recirculation. If RCS inventory decreases sufficiently to cause cold leg voiding, then steam flows through the cold leg balance lines to the CMTs.

Following an event such as steam-line break, the RCS experiences a decrease in temperature and pressure due to an increase of energy removed by the secondary system as a consequence of the break. The cooldown results in a reduction of the core shutdown margin due to the negative moderator temperature coefficient. There is a potential return to power, assuming the most reactive rod cluster control assembly is stuck in its fully withdrawn position. The actuation of the CMTs following this event provides injection of borated water via water recirculation to mitigate the reactivity transient and provide the required shutdown margin.

In case of a steam generator tube rupture, CMT injection together with the steam generator overfill prevention logic terminates the RCS leak into the steam generator. This occurs without actuation of the automatic depressurization system and without operator action. In a steam generator tube rupture, the CMTs operate in the water recirculation mode to provide borated water to compensate for RCS inventory losses and to borate the RCS. In case of an RCS leak rate of 10 gallons per minute, the PXS can delay the automatic depressurization system actuation for at least 10 hours while providing makeup water to the RCS. After the actuation of the automatic depressurization system, the PXS provides sufficient borated water to compensate for RCS shrinkage and to provide the RCS boration.

Samples from the CMTs are taken periodically in accordance with TS 3.5.2 surveillance requirements to check boron concentration of the solution in the CMT. Prior to the collection of liquid samples either in the laboratory or in the grab sampling unit, the lines are purged with source liquid to provide representative samples. The purging flow returns to the effluent holdup tank of the liquid radwaste system. Because the CMT is in open communication with the RCS via the balance line, whenever the surveillance is performed, the volume removed is inherently

replaced with RCS water via the balance line. The RCS water is at a lower boron concentration; therefore, each sample dilutes the CMT.

In order to increase CMT boron concentration, a fill connection is provided for CMT make up water from the chemical and volume control system (CVS) which is at a concentration > 3400 ppm. During this CVS make up activity (a feed-and-bleed process), the CMT volume displaced is added to the RCS via the balance line. This has the undesired effect of increasing the RCS boron concentration.

2.2 Current Licensing Basis Requirements

TS 3.5.2 LCO includes a Note stating that for Unit 4 until entry into TS 3.5.3 Applicability during shutdown for 4R01, CMT A may be considered OPERABLE with boron concentration ≤ 3400 ppm provided: (a) CMT A boron concentration is ≥ 3100 ppm and (b) the average of CMT A and CMT B boron concentration is ≥ 3400 ppm.

TS SR 3.5.2.4 requires verifying that the boron concentration in each CMT is ≥ 3400 ppm and ≤ 4500 ppm at a frequency of every 31 days. TS SR 3.5.2.4 includes a Note stating that this SR is not required to be performed for Unit 4 until entry into TS 3.5.3 Applicability during startup from 4R01.

UFSAR subsection 15.5.1, Inadvertent Operation of the Core Makeup Tanks During Power Operation, describes a scenario that would result from spurious CMT operation at power. The analysis assumes both CMTs inject 3400 ppm borated water, with the first CMT injecting at 10 seconds into the event and the second CMT injecting when the Low-2 T_{cold} "S" setpoint is reached 2,576.8 seconds later. The analysis concludes that inadvertent operation of the CMTs during power operation does not result in water being relieved from the pressurizer safety valves, departure from nucleate boiling (DNBR) remains above the design limit values, and the RCS and steam generator pressures remain below 110 percent of their design values.

2.3 Description of Proposed Change

A change is proposed to TS SR 3.5.2.4 to add a Note exception to the normal requirement to verify each CMT boron concentration is ≥ 3400 ppm, which will allow one CMT to meet its Surveillance (i.e., be considered OPERABLE) with boron concentration less 3400 ppm, provided that the CMT is ≥ 3100 ppm and that the average of both CMT boron concentrations is ≥ 3400 ppm. The specific wording of the Note to be added to TS SR 3.5.2.4 is:

> One CMT boron concentration may be < 3400 ppm provided it is ≥ 3100 ppm and the average of OPERABLE CMT boron concentrations is ≥ 3400 ppm.

The TS 3.5.2 LCO, which is applicable in Modes 1, 2 and 3, and Mode 4 with the RCS not being cooled by the Normal Residual Heat Removal System (RNS), specifies that "Both CMTs shall be OPERABLE." TS LCO 3.5.2 is revised to delete the Note stating:

> For Unit 4 until entry into TS 3.5.3 Applicability during shutdown for the first VEGP Unit 4 refueling outage (4R01), CMT A may be considered OPERABLE with boron concentration < 3400 ppm provided:
>
> a. CMT A boron concentration ≥ 3100 ppm and
>
> b. Average of CMT A and CMT B boron concentration is ≥ 3400 ppm.

TS SR 3.5.2.4 is revised to delete the Note stating that this SR is:

> Not required to be performed for Unit 4 until entry into TS 3.5.3 Applicability during startup from 4R01.

The analysis of the inadvertent CMT actuation is proposed to be revised to reflect the assumption that the inadvertently actuated CMT contains 3100 ppm of boron. With this revised assumption, the rate of core reactivity change will be different than the current analysis assuming 3400 ppm, and the automatic rod control response will be slightly different, but the plant would still be held to full power conditions until reactor trip. Because CMT boron is not required for event mitigation (i.e., the core will remain subcritical after reactor trip without crediting CMT boron), and because approximately the same conditions would be maintained by automatic rod control, there is no significant change in overall plant response to the event. A new LOFTRAN analysis was performed by Westinghouse to confirm this conclusion.

The updated time sequence of events results and event-related figures are provided in the markups to UFSAR subsection 15.5.1 in Attachment 1.

Markups showing the TS changes are also provided in Attachment 1. Clean typed TS change pages are shown in Attachment 2. Attachment 3 provides TS Bases markups consistent with the TS changes, for information only.

2.4 Reason for Proposed Change

Due to the arrangement of the CMTs (reference UFSAR Figure 6.3-1 Sheet 1), when borated water leaves a CMT, RCS is drawn from the RCS cold leg into the CMT through the balance line. When sampling CMTs (e.g., as required by SR 3.5.2.4) in-leakage from the RCS to the CMT occurs. CMT in-leakage can also occur as result CMT outlet valve leak-by. Because the RCS is at a lower boron concentration than the CMT, the in-leakage reduces the boron concentration in the CMT. In addition to sampling and in-leakage, the large concentration gradient between the RCS and the borated water in the CMT also results in some diffusion from the CMT through the normally open balance line and reduces the boron concentration in the CMT.

In August 2025, the Vogtle Unit 4 CMT “A” was experiencing CMT in-leakage both from sampling activities and due to leak-by of the CMT discharge isolation valves (PXS-PL-V014A / V015A). This prompted SNC’s request for an emergency license amendment on August 11, 2025 [ADAMS Accession No. ML25223A334], as supplemented by letter dated August 12, 2025 [ML25224A262]. At that time, Vogtle Unit 4 was at the end of cycle so the RCS boron concentration was at its lowest concentration, thereby creating a large concentration gradient between the RCS boron concentration and the CMTs which are connected to the RCS. This dynamic created challenges in maintaining CMT boron concentration.

NRC approved the LAR as VEGP Unit 4 Amendment No. 202 on August 15, 2025 [ML25225A229], which revised TS 3.5.2 to add a note to the (LCO) and another note in SR 3.5.2.4 regarding sampling for boron concentration. The amendment temporarily reduced the minimum allowable CMT A boron concentration and to not require performance of SR 3.5.2.4 until startup from the first Vogtle, Unit 4, refueling outage (RFO), which was scheduled for the fall of 2025. The amendment was a one-time temporary change that was effective until startup from the first Vogtle, Unit 4, RFO, which was completed in October 2025.

VEGP Units 3 and 4 operations have not been challenged to maintain CMT boron concentration since the CMT discharge valve leak-by that led to the requested VEGP Unit 4 Amendment. Furthermore, the CMT boron concentration levels are currently being maintained within TS limits, and no changes are proposed to the 31-day Frequency for this SR, which to date has been demonstrated to be sufficient during normal plant operating conditions. However, it is possible that a situation similar to that which led to the emergency amendment request could

reoccur. Therefore, to avoid reoccurrence of the need for another similar action, SNC is proposing this LAR to make permanent and applicable to both units' CMTs the minimum allowable CMT boron concentrations that were previously reviewed and approved for VEGP Unit 4 CMT A by VEGP Unit 4 Amendment No. 202.

Since the requested change would be permanent, the allowance that was included in the LCO Note is reformatted to an SR 3.5.2.4 Note as a more appropriate presentation consistent with the TS Writer's Guide, TSTF-GG-05-01. The proposed SR 3.5.2.4 Note is formatted similar to other SR Notes that acknowledge an exception or alternative acceptance criteria. For example, SR 3.2.3.3 provides an exception to the SR required determination by measurement of target flux difference; allowing for the initial target flux difference after each refueling to be determined based on design predictions. Another example is provided in SR 3.8.1.3, which allows a battery modified performance discharge test that satisfies SR 3.8.7.6 to be performed in lieu of SR 3.8.1.3. In these examples, each alternative acceptance criterion is stated similarly with a "may be" Note provision.

One additional difference in the proposed SR Note compared to the emergency LAR LCO Note is the addition of the requirement that the CMTs to be averaged are "OPERABLE" CMTs. This precludes satisfying the Note by averaging an inoperable CMT, e.g., one that is not meeting one of the other SRs, such as failing the SR 3.5.2.5 requirement that the CMT outlet isolation valve can stroke open, which would render the CMT incapable of meeting an assumption of the safety analysis.

Furthermore, because the changes approved by VEGP Unit 4 Amendment No. 202 were temporary and expired upon entry into TS 3.5.3 Applicability during shutdown for 4R01, SNC is proposing to remove the obsolete LCO 3.5.2 and SR 3.5.2.4 Notes, which were approved by this License Amendment.

3. TECHNICAL EVALUATION

Summary of Differences from Emergency LAR Evaluation

The emergency LAR submitted in August 2025 provided evaluations that were applicable to the emergent conditions under which that licensing request was made but were determined to be unnecessary for this non-emergent, standard license amendment request.

Because of the challenges with being at the end of fuel cycle and the valve leak-by conditions at that time, the site could not have reasonably foreseen the challenges to maintaining TS limits and steady-state operation over the final few weeks of the unit's first fuel cycle. Accordingly, consistent with 10 CFR 50.91(a)(5), the emergency LAR provided the basis for the request for emergency processing of the licensing request. As this LAR is not subject to the same emergency situation, in which failure to act in a timely way would result in derating or shutdown of a nuclear power plant, the basis for emergency processing provided in the emergency LAR is inapplicable to this LAR.

Additionally, the emergency LAR provided an engineering evaluation to confirm that CMT A met its defined safety functions and would continue to satisfy these safety functions throughout the remaining duration of the operating cycle, given the observed leakage from the affected CMT. This engineering evaluation is not applicable to the current LAR, because the emergent conditions under which the emergency LAR was submitted (i.e., CMT outlet valve leak-by and end of cycle conditions) are not currently present for either VEGP Unit 3 or Unit 4. Within this amendment request, the proposed minimum boron concentration limits are justified as

applicable at any time throughout the plant's operating cycle, as applied to the design basis events evaluated below, which are applicable throughout the operating cycle.

The engineering evaluation in the emergency LAR also considered the CMT B contribution to the 3400 ppm combined average concentration limit and reached the conclusion that there was reasonable assurance that CMT B would continue to support the average concentration limit until the LCO 3.5.2 applicability was exited in 4R01 without further sampling required by SR 3.5.2.4. As the current LAR does not request deferral of SR 3.5.2.4, as granted by the temporary amendment, this evaluation is no longer applicable.

The emergency LAR also provided a discussion of compensatory measures that would be implemented as a defense-in-depth effort to provide assurance that the valve leak-by would not cause the CMT A boron concentration to fall below the temporary limits during the period of applicability of this temporary LAR. Following approval of this permanent request, any results of sampling indicating a CMT boron concentration below the normal minimum limit (3400 ppm) would be identified as a deficient condition under the Corrective Action Program and continued operability would be assessed and maintained in accordance with established plant processes for operability determinations (which would include verification that the average of OPERABLE CMT boron concentrations is ≥ 3400 ppm). Compensatory measures, such as enhanced monitoring and trending, may be deemed appropriate to maintain the basis for the operability declaration or conformance with the TS, as documented in the Operability Determination Support Basis. Therefore, while the Adverse Condition Monitoring Plan (ACMP) was developed and implemented for the specific conditions summarized in the emergency LAR, any future deficient conditions related to the minimum CMT boron concentration limits would be subject to unique monitoring and trending activities determined by the issue-specific operability determinations.

For the emergency LAR discussion of the Anticipated Transient Without SCRAM (ATWS), a discussion of the end of cycle (EOC) moderator temperature coefficient (MTC) was included as a qualitative defense-in-depth evaluation to explain why ATWS was less limiting at the end of life. For this LAR, sensitivity cases were considered at various CMT boron concentrations, which determined that the acceptance criterion (3215 psia pressurizer limit) is maintained for both 3400 ppm nominal and 3100 ppm. These analyses credited the most limiting MTC across the entire cycle for the reanalysis. The most limiting MTC is at the start of the cycle and bounds the MTC at the end of the cycle. As such, this previous evaluation is not provided for the permanent change.

Design Basis Event Evaluation

The following non-LOCA events credit boration from CMT actuation:

- Steamline and Feedline Break (SLB/FLB) Mass and Energy (M&E) Release
- Hot Zero Power Steamline Break (HZP SLB): Core Response
- Increase in Reactor Coolant Inventory [Inadvertent CMT Actuation (ICMT) and Chemical and Volume Control (CVS) Malfunction (CVSM)]
- Loss of Normal Feedwater (LONF), Loss of ac Power (LOAC), and FLB (non-M&E)
- Safe Shutdown Duration (SSD) and Safe Shutdown Temperature (SST) non-Design Basis Accident (DBA) events
- Anticipated Transient Without SCRAM (ATWS) non-DBA events

Some non-LOCA events, including SLB/FLB M&E, HZP SLB, and CVSM, credit CMT injection with a minimum boron concentration of 3400 ppm in both CMTs. Since there is no credible single failure that would prevent a CMT from actuating based on having CMT outlet valves in parallel, as long as the average boron concentration from both CMTs is ≥ 3400 ppm, these non-LOCA events as analyzed in Chapters 6 and 15 of the UFSAR would not be impacted.

The ICMT non-LOCA event models a single CMT actuating inadvertently as the initiating event. To meet the primary acceptance criterion for the ICMT event, it is verified that the pressurizer does not become filled with water. The pressurizer fills with water as a result of CMT injection; therefore, the CMT volume, and not the negative reactivity contribution of CMT boration, is a major contributor to the analysis acceptance criterion. This is confirmed in UFSAR subsection 15.5.1.1, which states that the reactor is tripped on a high-3 pressurizer level signal and does not credit boron injected from the CMT to shut down the core or to prevent a return to power in the analysis. However, the change to the faulted CMT's boron concentration results in minor changes to the automatic rod control response prior to the reactor trip. UFSAR markups reflecting revisions to the event response based on faulted CMT revised boron concentration of 3100 ppm are provided in Attachment 1. The faulted CMT boron concentration change does not impact the overall conclusions of the analysis; the revised analysis verifies that the pressurizer is not overfilled beyond the acceptable limit.

Other non-LOCA events model the CMTs at a maximum boron concentration of 4500 ppm. For the LONF, LOAC, and FLB events, a higher boron concentration was found to be bounding. As such, this change to the minimum CMT boron concentration would not adversely impact the LONF, LOAC, or FLB non-M&E events.

The non-DBA ATWS scenario discussed in UFSAR Section 15.8 is associated with demonstrating compliance to 10 CFR 50.62 (ATWS Rule). For the ATWS analyses, CMT boron concentration helps control the power excursion thereby limiting the RCS pressure increase. Changes to the non-LOCA ATWS analyses could impact the probabilistic risk assessment (PRA) core damage frequency (CDF). ATWS analyses model a nominal CMT boron concentration that is slightly above the minimum average CMT boron concentration limit. Updated LOFTRAN cases were conservatively analyzed with a minimum CMT boron concentration of 3400 ppm for cases modeling both CMTs and 3100 ppm for cases modeling only one CMT. These cases demonstrated that sufficient margin remains for the ATWS reactor coolant system pressure limit (3215 psia); therefore, there is no adverse impact on any ATWS cases. Consequently, there is no impact on the 10 CFR 50.62 CDF safety goals.

The non-DBA SSD/SST analyses also model a nominal CMT boron concentration that is slightly above the minimum average CMT boron concentration limit. During the event, the CMTs actuate after the reactor trip; as such, CMT boron has a negligible impact on the results of these analyses so long as the core remains subcritical. The transient results confirm that the core remains subcritical. Therefore, this event is not impacted by a change in the minimum boron concentration of a single CMT when the average of both CMTs remains unchanged.

The UFSAR Chapter 15 evaluations for the following LOCA events do not model boron concentration:

- Steam Generator Tube Rupture (SGTR)
- Small Break Loss of Coolant Accident (SBLOCA)
- Large Break Loss of Coolant Accident (LBLOCA)

In the case of an SGTR, CMT injection provides borated water to compensate for RCS leakage; however, CMT boration is not credited in the SGTR event as control and shutdown rods are assumed to insert and provide adequate shutdown margin (SDM).

SBLOCA and LBLOCA analyses, as documented in UFSAR Section 15.6, demonstrate 10 CFR 50.46 compliance by crediting a reactor trip for maintaining shutdown margin. The SBLOCA and LBLOCA analyses do not model any potential return to power that require CMT boron concentration to prevent. Therefore, these analyses do not model CMT boron concentration as an input parameter and there is no adverse impact on the SBLOCA or LBLOCA analyses as a result of this change.

A separate analysis is performed for determining SDM provided by the emergency core cooling system (ECCS) for the LOCA events (SBLOCA and LBLOCA). The double-ended direct vessel injection (DEDVI) line break is the most limiting for determining SDM. Due to the break location for this event, one train of ECCS (one CMT, one Accumulator) is assumed to not inject into the reactor vessel. This limiting scenario was evaluated assuming the lower boron concentration CMT was the injection source. The results of this evaluation show that if the injected CMT was at a boron concentration of 3100 ppm there would be > 800 ppm of shutdown margin available. As such, the results of the 10 CFR 50.46 compliant analyses documented in Section 15.6 of the UFSAR are not impacted by this change.

In conclusion, Westinghouse has evaluated the impact of allowing either CMT A or CMT B boron concentration to be as low as 3100 ppm when performing its design function as discussed in UFSAR Chapters 6 and 15. The ability of the CMTs to provide adequate core cooling flow during injection and recirculation modes is not impacted by this change. The function of CMT boron to provide adequate SDM remains for either CMT A or CMT B boron concentration no lower than 3100 ppm provided the combined boron concentration from CMT A and CMT B is ≥ 3400 ppm.

4. REGULATORY EVALUATION

4.1 Applicable Regulatory Requirements/Criteria

This activity involves changes to the operating license Appendix A, Technical Specifications; therefore, in accordance with 10 CFR 50.90, this activity requires an amendment. As such, NRC approval is required prior to making the changes proposed in this license amendment request.

10 CFR 50, Appendix A, General Design Criterion (GDC) 26 requires, in part, that “[t]wo independent reactivity control systems of different design principles shall be provided.” The second reactivity control system is chemical shim (boric acid). The proposed change to reduce one CMT lower boron concentration provided the average of both CMT boron concentrations is satisfied is conservative as analyses have shown that sufficient boron concentration will be available to perform the design basis function assumed for the CMTs for credited design basis accidents. Therefore, the proposed changes comply with the requirement of GDC 26.

10 CFR Part 50, Appendix A GDC 27, “Combined reactivity control systems capability,” requires that “[t]he reactivity control systems be designed to have a combined capability, in conjunction with poison addition by the emergency core cooling system, of reliably controlling reactivity changes to assure that under postulated accident conditions and with appropriate margin for stuck rods the capability to cool the core is maintained.” The proposed change does not affect the means of making and holding the core subcritical under anticipated conditions and with

appropriate margin for contingencies. Shutdown margin is not affected. Therefore, the proposed changes comply with the requirement of GDC 27.

10 CFR Part 50, Appendix A GDC 35, “Emergency core cooling,” requires a system to provide abundant emergency core cooling whose safety function is to transfer heat from the reactor core following any loss of reactor coolant at a rate such that fuel and clad damage that could interfere with continued effective core cooling is prevented and clad metal-water reaction is limited to negligible amounts. The proposed change to the SR 3.5.2.4 does not change the design of the system to perform these functions with abundant emergency core cooling. Therefore, the proposed changes comply with the requirement of GDC 35.

10 CFR Part 50, Appendix A GDC 37 requires, in part, that the emergency core cooling system be designed to permit appropriate periodic functional testing to assure the operability of the system. The approved periodic functional testing of the CMT boron concentration to confirm the operability of the system remains unchanged.

10 CFR 50.36(c)(3) requires that TSs include Surveillance Requirements. Per 10 CFR 50.36(c)(3), SRs “are requirements relating to test, calibration, or inspection to assure that the necessary quality of systems and components is maintained, that facility operation will be within safety limits, and that the limiting conditions for operation will be met.” The proposed change to SR 3.5.2.4 to reduce the minimum allowable boron concentration of the CMTs continues to reflect the lowest functional capability required for safe operation and continues to provide appropriate remedial actions including a required plant shutdown if they are not met.

4.2 Precedent

The proposed change makes permanent a temporary change to the VEGP Unit 4 operating license, Appendix A, TS 3.5.2, Core Makeup Tanks (CMTs) - Operating, SR 3.5.2.4 to reduce the minimum allowable CMT boron concentration, as approved by VEGP Unit 4 Amendment No. 202, on August 15, 2025 [ADAMS Accession No. ML25225A229].

4.3 Significant Hazards Consideration

In accordance with 10 CFR 50.90, Southern Nuclear Operating Company (SNC) hereby requests a license amendment to the Technical Specifications (TS) for Vogtle Electric Generating Plant (VEGP) Units 3 and 4 (Combined Licenses (COLs) NPF-91 and NPF-92, respectively). This license amendment request (LAR) proposes change to the COLs Appendix A, TS to revise TS 3.5.2, Core Makeup Tanks (CMTs) - Operating, Surveillance Requirement (SR) 3.5.2.4, to make permanent and applicable to both CMTs for both units the minimum allowable CMT boron concentration and the minimum average CMT boron concentration that was previously approved for VEGP Unit 4 [ADAMS Accession No. ML25225A229]. This request also proposes changes to delete the temporary Notes that were added to Limiting Condition for Operation (LCO) 3.5.2 and SR 3.5.2.4 by the previously approved amendment for VEGP Unit 4. To support the CMT boron concentration limit change, this request also presents a revision to the analysis of the inadvertent operation of the CMTs during power operation event, as discussed in Updated Final Safety Analysis Report (UFSAR) subsection 15.5.1.

SNC has evaluated whether a significant hazards consideration is involved with the proposed amendment(s) by focusing on the three standards set forth in 10 CFR 50.92, "Issuance of amendment," as discussed below:

1) Does the proposed amendment involve a significant increase in the probability or consequences of an accident previously evaluated?

Response: No.

The proposed changes do not adversely affect the operation of any structures, systems, or components (SSCs) associated with an accident initiator or initiating sequence of events. The proposed changes do not affect the design of the passive core cooling system.

The proposed amendment does not affect accident initiators or precursors nor adversely alter the design assumptions, conditions, or configuration of the facility. The proposed amendment does not alter any plant equipment or operating practices with respect to such initiators or precursors in a manner that the probability of an accident is increased. The proposed amendment to reduce the minimum allowable CMT boron concentration does not adversely affect the operation of the assumed mitigation systems or the containment fission product barrier assumptions. The requested reduction in minimum allowable boron concentration in one CMT is offset by the requirement to meet the minimum boron concentration average of both CMTs. As such, the proposed change will not alter assumptions relative to the mitigation of an accident or transient event. The proposed amendment does not increase the likelihood of the malfunction of a structure, system, or component (SSC) or adversely impact analyzed accidents.

Therefore, the proposed change does not involve a significant increase in the probability or consequences of an accident previously evaluated.

2) Does the proposed amendment create the possibility of a new or different kind of accident from any accident previously evaluated?

Response: No.

The proposed changes do not change the design function of the CMTs. These proposed changes do not introduce any new equipment or components that would result in a new failure mode, malfunction or sequence of events that could adversely affect safety-related or non-safety-related equipment. This activity will not allow for a new fission product release path, result in a new fission product barrier failure mode, or create a new sequence of events that would result in significant fuel cladding failures.

Therefore, the proposed change does not create the possibility of a new or different kind of accident from any accident previously evaluated.

3) Does the proposed amendment involve a significant reduction in a margin of safety?

Response: No.

The margin of safety is related to the ability of the CMTs to provide adequate shutdown margin as assumed in safety analyses. Evaluation of the impact of allowing the boron concentration of a single CMT to be less than the required limit on the capability of the passive core cooling system to perform its design function has confirmed that the safety analyses assumptions regarding CMT boron concentration are bounded provided the average boron concentration from both CMTs is greater than the proposed combined average limit. No safety analysis or design basis acceptance limit/criterion is challenged or exceeded by the requested change, thus no margin of safety is reduced.

Therefore, the proposed change does not involve a significant reduction in a margin of safety.

Based on the above, SNC concludes that the proposed amendment does not involve a significant hazards consideration under the standards set forth in 10 CFR 50.92(c), and, accordingly, a finding of "no significant hazards consideration" is justified.

4.4 Conclusions

In conclusion, based on the considerations discussed above, (1) there is reasonable assurance that the health and safety of the public will not be endangered by operation in the proposed manner, (2) such activities will be conducted in compliance with the Commission's regulations, and (3) the issuance of the amendment will not be inimical to the common defense and security or to the health and safety of the public.

5. ENVIRONMENTAL CONSIDERATION

The proposed changes to the Technical Specifications (TS) are described in Section 2 of this Enclosure, which require a license amendment. SNC has evaluated this license amendment request against the criteria for identification of licensing and regulatory actions requiring environmental assessment in accordance with 10 CFR 51.21.

The requested amendment meets the eligibility criteria for categorical exclusion set forth in 10 CFR 51.22(c)(9), in that the amendment would not involve (i) a significant hazards consideration, (ii) a significant change in the types or significant increase in the amounts of any effluents that may be released offsite, or (iii) a significant increase in individual or cumulative occupational radiation exposure.

Accordingly, the proposed amendment meets the eligibility criteria for categorical exclusion set forth in 10 CFR 51.22(c)(9). Therefore, pursuant to 10 CFR 51.22(b), no environmental impact statement or environmental assessment need be prepared in connection with the proposed amendment.

Attachment 1
to the Enclosure of NL-25-0455

Proposed Technical Specification and Updated Final Safety Analysis Report (UFSAR) Changes (Mark-ups)

Insertions Denoted by underlined Blue text and Deletions by ~~Red strikethrough~~.

3.5 PASSIVE CORE COOLING SYSTEM (PXS)

3.5.2 Core Makeup Tanks (CMTs) – Operating

LCO 3.5.2 Both CMTs shall be OPERABLE.

~~**- NOTE -**~~

~~For Unit 4 until entry into TS 3.5.3 Applicability during shutdown for 4R01, CMT A may be considered OPERABLE with boron concentration < 3400 ppm provided:~~

~~a. CMT A boron concentration is ≥ 3100 ppm and~~

~~b. Average of CMT A and CMT B boron concentration is ≥ 3400 ppm.~~

APPLICABILITY: MODES 1, 2, and 3.
MODE 4 with the Reactor Coolant System (RCS) not being cooled by the Normal Residual Heat Removal System (RNS).

ACTIONS

CONDITION	REQUIRED ACTION	COMPLETION TIME
A. One CMT inoperable due to one CMT outlet isolation valve inoperable.	A.1 Restore outlet isolation valve to OPERABLE status.	72 hours
B. One CMT inoperable due to water temperature or boron concentration not within limits.	B.1 Restore water temperature and boron concentration to within limits.	72 hours
C. Two CMTs inoperable due to water temperature or boron concentration not within limits.	C.1 Restore water temperature and boron concentration to within limits for one CMT.	1 hour from discovery of LCO 3.5.2 Condition C entry concurrent with LCO 3.5.1 Condition B entry <u>AND</u> 8 hours

SURVEILLANCE REQUIREMENTS (continued)

SURVEILLANCE		FREQUENCY
SR 3.5.2.4	---------- **- NOTE -** ~~Not required to be performed for Unit 4 until entry into TS 3.5.3 Applicability during startup from 4R01.~~ One CMT boron concentration may be < 3400 ppm provided it is ≥ 3100 ppm and the average of OPERABLE CMT boron concentrations is ≥ 3400 ppm. ---------- Verify the boron concentration in each CMT is ≥ 3400 ppm and ≤ 4500 ppm.	31 days
SR 3.5.2.5	Verify each CMT outlet isolation valve strokes open.	In accordance with the Inservice Testing Program
SR 3.5.2.6	Verify each CMT outlet isolation valve actuates to the open position on an actual or simulated actuation signal.	24 months
SR 3.5.2.7	Verify system flow performance of each CMT in accordance with the System Level OPERABILITY Testing Program.	10 years

- Two out of four Low-2 steam line pressure signals in any one loop (above permissive P-11)
- Two out of four high negative steam pressure rates in any one loop (below permissive P-11)
- Two out of four Low-2 T_{cold} signals in any one loop, or
- Two out of four high-2 containment pressure signals

Plant systems and equipment available to mitigate the effects of the accident are discussed in Subsection 15.0.8 and listed in Table 15.0-6.

15.1.4.2 Analysis of Effects and Consequences

15.1.4.2.1 Method of Analysis

The analysis of a secondary system steam release is performed to determine the following:

- The core heat flux and reactor coolant system temperature and pressure resulting from the cooldown, due to the steam release. The LOFTRAN code (References 1 and 6) is used to model the system transient.

- The thermal-hydraulic behavior of the core due to the steam release. A detailed thermal-hydraulic digital computer code, VIPRE-01 (Reference 7), is used to determine if DNB occurs for the core transient conditions computed by the LOFTRAN code.

The following conditions are assumed to exist at the time of a secondary system steam release:

- End-of-life shutdown margin at no-load, equilibrium xenon conditions, and with the most reactive RCCA stuck in its fully withdrawn position. Operation of RCCA mechanical shim and axial offset banks during core burnup is restricted by the insertion limits so that shutdown margin requirements are satisfied.

- A most negative moderator temperature coefficient corresponding to the end-of-life rodded core with the most reactive RCCA in the fully withdrawn position. The variation of the coefficient with temperature is included. The k_{eff} (considering moderator temperature and density effects) versus temperature corresponding to the negative moderator temperature coefficient used is shown in Figure 15.1.4-1. The core power is calculated as a function of core mass flow, core boron concentration, and core inlet temperature.

- Minimum capability for injection of boric acid solution corresponding to the most restrictive single failure in the passive core cooling system. There are no single failures that prevent core makeup tank injection, however, the analysis models the failure of one core makeup tank discharge valve. Low-concentration boric acid must be swept from the core makeup tank lines downstream of isolation valves before delivery of boric acid ~~(3400 ppm)~~ to the reactor coolant loops. This effect has been accounted for in the analysis.

- The case analyzed models a flow area of 0.2 ft^2, which is based on a steam flow of 520 pounds per second at 1200 psia with offsite power available. This conservatively bounds the maximum capacity of any single steam dump, relief, or safety valve.

- Initial hot shutdown conditions at time zero are assumed because this represents the most conservative initial conditions. Should the reactor be just critical or operating at power at the time of a steam release, the reactor is tripped by the normal overpower protection when power level reaches a trip point. Following a trip at power, the reactor coolant system contains more stored energy than at no-load. This is because the average coolant

temperature is higher than at no-load, and there is appreciable energy stored in the fuel. The additional stored energy is removed via the cooldown caused by the steam release before the no-load conditions of the reactor coolant system temperature and shutdown margin assumed in the analyses are reached. After the additional stored energy is removed, the cooldown and reactivity insertions proceed in the same manner as in the analysis that assumes no-load condition at time zero. However, because the initial steam generator water inventory is greatest at no-load, the magnitude and duration of the reactor coolant system cooldown are less for a steam line release occurring at power:

- In computing the steam flow, the Moody Curve (Reference 3) for f(L/D) = 0 is used.
- Perfect moisture separation occurs in the steam generator.
- Offsite power is available, because this maximizes the cooldown.
- Maximum cold startup feedwater flow is assumed.
- Four reactor coolant pumps are initially operating.
- Manual actuation of the PRHR system at time zero is conservatively assumed to maximize the cooldown.

15.1.4.2.2 Results

The calculated sequence of events for the analyzed case is shown in Table 15.1.2-1. The results presented conservatively indicate the events that would occur assuming a secondary system steam release because it is postulated that the conditions described in Subsection 15.1.4.2.1 exist simultaneously.

an average of 3400 ppm from both CMTs

Figures 15.1.4-2 through 15.1.4-12 show the transient results for the event. The steam release accounted for in the analysis is bounding compared to the capacity of any single steam dump, relief, or safety valve.

Core makeup tank injection and the associated tripping of the reactor coolant pumps are initiated automatically by the Low-2 T_{cold} "S" signal. Boron solution at ~~3400 ppm~~ enters the reactor coolant system, providing enough negative reactivity to prevent a significant return to power and core damage. Later in the transient, as the reactor coolant pressure continues to fall, the accumulators actuate and inject boron solution at 2600 ppm.

The transient is conservative with respect to cooldown, because no credit is taken for the energy stored in the system metal other than that of the fuel elements and steam generator tubes, and the PRHR system is assumed to be actuated at time zero. Because the limiting portion of the transient occurs over a period of about 5 minutes, the neglected stored energy would have a significant effect in slowing the cooldown.

The calculated time sequence of events for this accident is listed in Table 15.1.2-1.

15.1.4.3 Margin to Critical Heat Flux

The analysis demonstrates that the DNB design basis, as described in Section 4.4, is met for the inadvertent opening of a steam generator relief or safety valve. As shown in Figure 15.1.4-2, no significant return to power occurs and, therefore, DNB does not occur. The minimum DNBR is conservatively calculated and is above the 95/95 limit.

- The core heat flux and reactor coolant system temperature and pressure resulting from the cooldown following the steam line break. The LOFTRAN code (References 1 and 6) is used to model the system transient.

- The thermal-hydraulic behavior of the core following a steam line break. A detailed thermal-hydraulic digital computer code, VIPRE-01 (Reference 7), is used to determine if DNB occurs for the core transient conditions computed by the LOFTRAN code.

The following conditions are assumed to exist at the time of a main steam line break accident:

- End-of-cycle shutdown margin at no-load, equilibrium xenon conditions, and the most reactive RCCA stuck in its fully withdrawn position. Operation of RCCA mechanical shim and axial offset banks during core burnup is restricted by the insertion limits so that shutdown margin requirements are satisfied.

- A most negative moderator temperature coefficient corresponding to the end-of-life rodded core with the most reactive RCCA in the fully withdrawn position. The variation of the coefficient with temperature is included. The k_{eff} (considering moderator temperature and density effects) versus temperature corresponding to the negative moderator temperature coefficient used is shown in Figure 15.1.4-1. The core power is calculated as a function of core mass flow, core boron concentration, and core inlet temperature.

The moderator properties used in the LOFTRAN code for feedback calculations are generated by combining those in the sector nearest the affected steam generator with those associated with the remaining sector. The resultant properties reflect a combination process that accounts for inlet plenum fluid mixing and a conservative weighing of the fluid properties from the coldest core sector.

In verifying the conservatism of this method, the power predictions of the LOFTRAN modeling are confirmed by comparison with detailed core analysis for the limiting conditions of the cases considered. This core analysis conservatively models the hypothetical core configuration (that is, stuck RCCA, non-uniform inlet temperatures, pressure, flow, and boron concentration) and directly evaluates the total reactivity feedback including power, boron, and density redistribution in an integral fashion. The effect of void formation is also included.

Comparison of the results from the detailed core analysis with the LOFTRAN predictions verifies the overall conservatism of the methodology. That is, the specific power, temperature, and flow conditions used to perform the DNB analysis are conservative.

- Minimum capability for injection of boric acid solution corresponding to the most restrictive single failure in the passive core cooling system. The core makeup tanks and the accumulators are the portions of the passive core cooling system used in mitigating a steam line rupture. There are no single failures that prevent core makeup tank injection however, the analysis models the failure of one core makeup tank discharge valve. Low-concentration boric acid must be swept from the core makeup tank lines downstream of isolation valves before delivery of boric acid ~~(3400 ppm)~~ to the reactor coolant loops. This effect has been accounted for in the analysis.

- The maximum overall fuel-to-coolant heat transfer coefficient is used to maximize the rate of cooldown.

- Because the steam generators are provided with integral flow restrictors with a 1.4-ft^2 throat area, any rupture in a steam line with a break area greater than 1.4 ft^2, regardless of location, has the same effect on the primary plant as the 1.4-ft^2 double-ended rupture. The limiting case considered in determining the core power and reactor coolant system transient is the

- Four reactor coolant pumps are initially operating.
- Manual actuation of the PRHR system at time zero is conservatively assumed in order to maximize the cooldown.

15.1.5.2.2 Results

The calculated sequence of events for the analyzed case is shown in Table 15.1.2-1. The results presented conservatively indicate the events that would occur assuming a steam line rupture because it is postulated that the conditions described in Subsection 15.1.5.2.1 exist simultaneously.

15.1.5.2.3 Core Power and Reactor Coolant System Transient

Figures 15.1.5-1 through 15.1.3-13 show the transient results following a main steam line rupture (complete severance of a pipe) at initial no-load condition.

Offsite power is assumed available so that, initially, full reactor coolant flow exists. During the course of the event, the reactor protection system initiates a trip of the reactor coolant pumps in conjunction with actuation of the core makeup tanks. The transient shown assumes an uncontrolled steam release from only one steam generator. Steam release from more than one steam generator is prevented by automatic trip of the main steam isolation valves in the steam lines by Low-2 steam line pressure signals. Even with the failure of one valve, release is limited to approximately 10 seconds for the other steam generator while the one generator blows down. The main steam isolation valves fully close in less than 10 seconds from receipt of a closure signal.

at an average of

As shown in Figure 15.1.5-1, the core attains criticality with the RCCAs inserted (with the design shutdown assuming the most reactive RCCA stuck) before boron solution at 3400 ppm (from core makeup tanks) or 2600 ppm (from accumulators) enters the reactor coolant system. A peak core power significantly lower than the nominal full-power value is attained.

both

The calculation assumes that the boric acid is mixed with and diluted by the water flowing in the reactor coolant system before entering the reactor core. The concentration after mixing depends upon the relative flow rates in the reactor coolant system and from the core makeup tanks or accumulators (or both). The variation of mass flow rate in the reactor coolant system due to water density changes is included in the calculation. The variation of flow rate from the core makeup tanks or accumulators (or both) due to changes in the reactor coolant system pressure and temperature and the pressurizer level is also included. The reactor coolant system and passive injection flow calculations include line losses.

At no time during the analyzed steam line break event does the core makeup tank level approach the setpoint for actuation of the automatic depressurization system. During non-LOCA events, the core makeup tanks remain filled with water. The volume of injection flow leaving the core makeup tank is offset by an equal volume of recirculation flow that enters the core makeup tanks via the reactor coolant system cold leg balance lines.

The PRHR system provides a passive, long-term means of removing the core decay and stored heat by transferring the energy via the PRHR heat exchanger to the in-containment refueling water storage tank (IRWST). The PRHR heat exchanger is normally actuated automatically when the steam generator level falls below the Low-2 steam generator wide range water level. For the main steam line rupture case analyzed, the PRHR exchanger is conservatively actuated at time zero to maximize the cooldown.

- Control systems

  The pressurizer spray, the pressurizer heaters, and the automatic rod control system are conservatively assumed to operate. The pressurizer heaters are automatically blocked on a high-3 pressurizer level signal following a 15-second timer delay, but add heat to the system during the initial part of the transient. Other control systems are conservatively not assumed to function during the transient.

- Moderator and Doppler coefficients of reactivity

  A least-negative moderator temperature coefficient, a low (absolute value) Doppler power coefficient, and a minimum boron worth are assumed. With these minimum feedback parameters and the operability of the pressurizer spray system and automatic rod control system assumed, the reactivity effects of the boron injection from the core makeup tanks is counteracted. As a result, the High-3 pressurizer water level signal is the first reactor trip signal generated during the transient.

- Boron injection

  3100

  The transient is initiated by an inadvertent opening of the discharge valves of one of the two core makeup tanks, resulting in the injection of ~~3400~~ ppm borated water.

- Protection and safety monitoring system actuations

  The operators are assumed to be alerted to the pressurizer level increase transient by the high-2 pressurizer level signal. Reactor trip is initiated by the first of two high-3 pressurizer level signals. The second high-3 pressurizer level signal triggers the operators to open the reactor vessel head vent; this action is at least 30 minutes after the operator has been alerted by the high-2 pressurizer level signal.

  The core decay heat is removed by the PRHR heat exchanger. The worst single failure is assumed to occur in the outlet line of the PRHR heat exchanger. One of the two parallel isolation valves is assumed to fail to open.

Plant systems and equipment available to mitigate the effect of the accident are discussed in Subsection 15.0.8 and listed in Table 15.0-6.

15.5.1.3 Results

788.7

2,366.5

2,376.5

Figures 15.5.1-1 through 15.5.1-8 show the transient response to the inadvertent operation of one of the two core makeup tanks during power operation. The inadvertent opening of the core makeup tank discharge valves occurs at 10 seconds. As the core makeup tank adds inventory to the primary system, the pressurizer level begins to increase until the high-2 pressurizer level setpoint is reached (~~783.2~~ seconds) and continues until the high-3 pressurizer level reactor trip setpoint is reached at about ~~2,402.0~~ seconds. After a 2-second delay, the neutron flux starts decreasing due to the reactor trip, which is followed by turbine trip after a 5-second turbine trip delay. Following reactor trip, the reactor power drops and the average reactor coolant system temperature decreases with subsequent coolant shrinkage. Due to the assumed loss of offsite power, the reactor coolant pumps trip at about ~~2,412.0~~ seconds. The cold leg temperature increases and the core makeup tank starts injecting cold water into the reactor coolant system at a higher rate due to the increased driving head resulting from the density decrease in the balance line and due to the reduced pressure drop between the cold leg and the injection line connection on the reactor vessel following the trip of the reactor coolant pumps. The post-trip primary coolant system shrinkage is counteracted by the core makeup tank injection, and the pressurizer water volume starts to increase because of the heatup of

the cold injected fluid by the decay heat. The high-3 pressurizer level setpoint is once again reached at ~~2,554.8~~ 2,515.8 seconds, and after a 15-second timer delay, the signal is sent to actuate the PRHR heat exchanger and block the pressurizer heaters. Following a conservative 17-second delay, the valves are assumed to open to actuate the PRHR heat exchanger at ~~2,586.8~~ 2,547.8 seconds.

If the PRHR heat exchanger coolant asymmetrically injects into the steam generator outlet plenum, then one cold leg may reach the Low-2 T_{cold} "S" setpoint more quickly than if the flow were split evenly. To conservatively account for this effect, the Low-2 T_{cold} "S" signal is modeled to actuate simultaneously with the actuation of the PRHR heat exchanger (~~2,586.8~~ 2,547.8 seconds). The Low-2 T_{cold} "S" signal activates the second core makeup tank, which then begins injecting additional mass into the reactor coolant system. An earlier actuation of the second core makeup tank results in a more limiting pressurizer fill transient.

As the second core makeup tank begins injecting, the pressurizer level continues to increase. The operators are assumed to be alerted by the high-2 pressurizer level signal (~~783.2 seconds~~ 788.7) that a pressurizer level increase transient is underway, and it is assumed that the operators are ready to take corrective action no sooner than 30 minutes later. In this analysis, since pressurizer level continues to increase, the high-3 pressurizer level reactor trip setpoint is reached within this time. The operator action assumed in this case is to open the reactor vessel head vent to preclude overfill following receipt of the second high-3 pressurizer level signal (~~3,483.2~~ 3,488.7 seconds); this action is at least 30 minutes (45 minutes is conservatively assumed in the analysis) after the operator has been alerted by the high-2 pressurizer level signal.

The safety related reactor vessel head vent is opened by the operators and the pressurizer water level increase slows and eventually the level begins to decrease. This demonstrates that the capacity of the reactor vessel head vent is sufficient to preclude pressurizer overfill as a result of an inadvertent actuation of a core makeup tank.

During the event, the departure from nucleate boiling ratio (DNBR) would not drop significantly below the initial value due to the addition of highly borated water from the core makeup tanks to the reactor coolant system. At the time of reactor trip core power and heat flux drop rapidly, which would cause the DNBR to increase well above the design limit value defined in Section 4.4.

The calculated sequence of events is shown in Table 15.5-1.

As noted above, the limiting case presented here models explicit operator action 45 minutes after receipt of the high-2 pressurizer level signal. For pressurizer level increase events, the operator would take action to reduce the increase in coolant inventory. As the pressurizer water level would increase above the High-2 pressurizer water level that normally isolates chemical and volume control system makeup, the normal letdown line could be placed into service to reduce the increase in coolant inventory. If letdown could not be placed into service, the operator could use the safety related reactor vessel head vent valves to reduce the increase in coolant inventory (this is explicitly modeled in the case presented here). For these events, there is sufficient time for the operator to mitigate the consequences by following the procedures outlined in the Emergency Response Guidelines AFR-I.1.

15.5.1.4 Conclusions

The results of this analysis show that inadvertent operation of the core makeup tanks during power operation does not adversely affect the core, the reactor coolant system, or the steam system. Water is not relieved from the pressurizer safety valves. DNBR always remains above the design limit values, and reactor coolant system and steam generator pressures remain below 110 percent of their design values.

Table 15.5-1 (Sheet 1 of 2)
Time Sequence of Events For Incidents Which Result in an Increase in Reactor Coolant Inventory

Accident	Event	Time (seconds)
Inadvertent operation of the core makeup tanks during power operation	Core makeup tank discharge valves open	10.0
	High-2 pressurizer level setpoint reached	~~783.2~~ 788.7
	High-3 pressurizer level setpoint reached	~~2402.0~~ 2366.5
	Rod motion begins	~~2404.0~~ 2368.5
	Loss of offsite power	~~2412.0~~ 2376.5
	Reactor coolant pumps trip	~~2412.0~~ 2376.5
	High-3 pressurizer level setpoint reached	~~2554.8~~ 2515.8
	Pressurizer heaters blocked on high-3 level	~~2569.8~~ 2530.8
	PRHR heat exchanger actuated	~~2586.8~~ 2547.8
	Low-2 T_{cold} "S" setpoint is reached	~~2586.8~~ 2547.8
	Second CMT starts recirculating	~~2586.8~~ 2547.8
	Main steam and feed lines are isolated	~~2598.8~~ 2559.8
	Operators open the reactor vessel head vent after the high-3 pressurizer level signal is reached (at least 30 minutes after high-2 pressurizer level setpoint is reached; 45 min as analyzed)	~~3483.2~~ 3488.7
	Peak pressurizer water volume occurs	~~5312.5~~ 5242.5

Delete current UFSAR Figure 15.5.1-1:

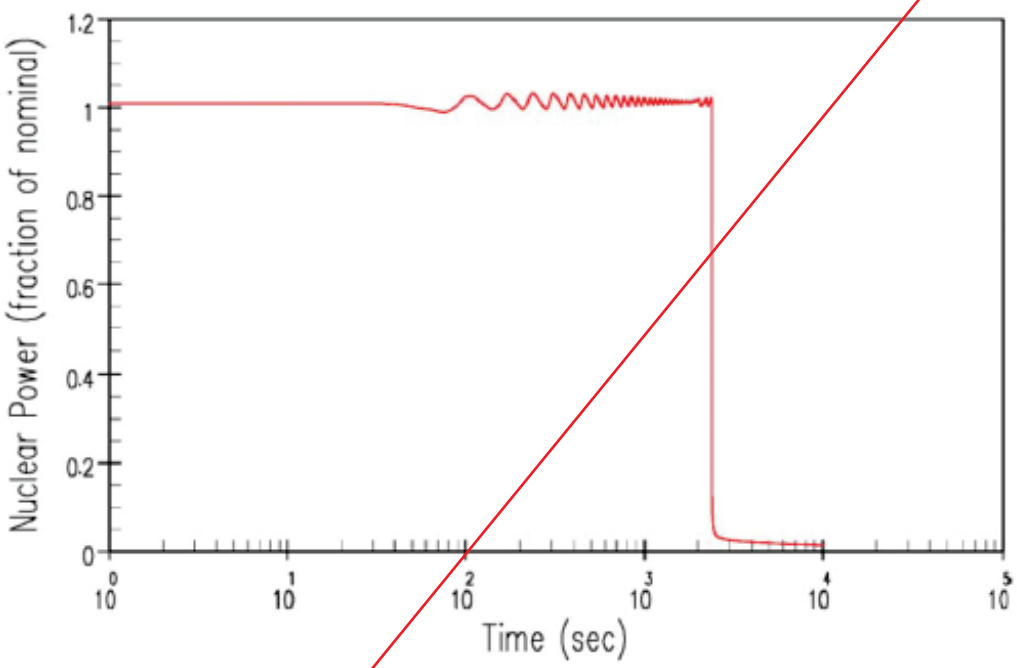
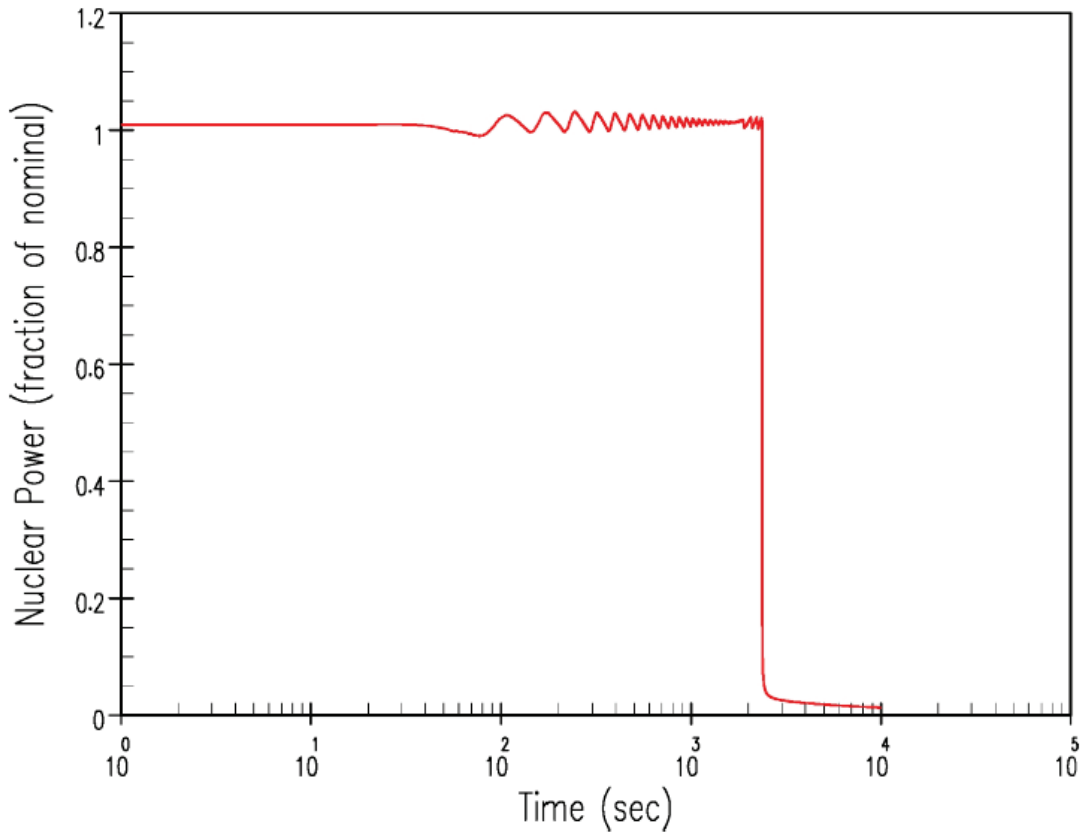


Figure 15.5.1-1
Core Nuclear Power Transient for Inadvertent Operation of the Emergency Core Cooling System Due to a Spurious Opening of the Core Makeup Tank Discharge Valves

Revised UFSAR Figure 15.5.1-1:



Updated UFSAR Figure 15.5.1-1
Core Nuclear Power Transient for Inadvertent Operation of the Emergency Core Cooling System Due to a Spurious Opening of the Core Makeup Tank Discharge Valves

Delete current UFSAR Figure 15.5.1-2:

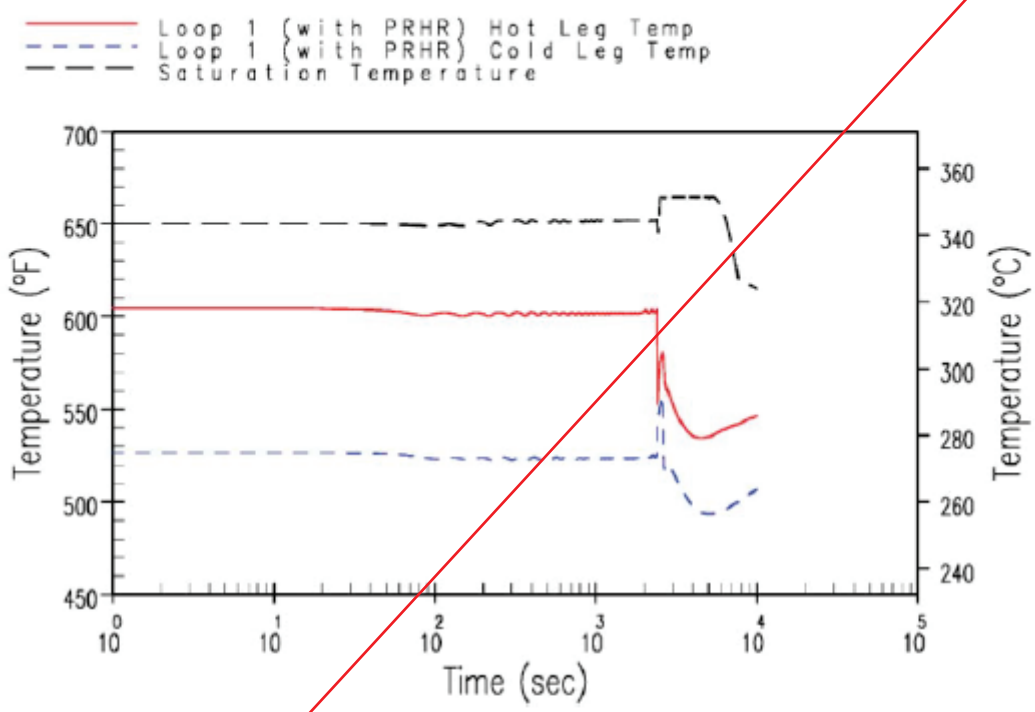
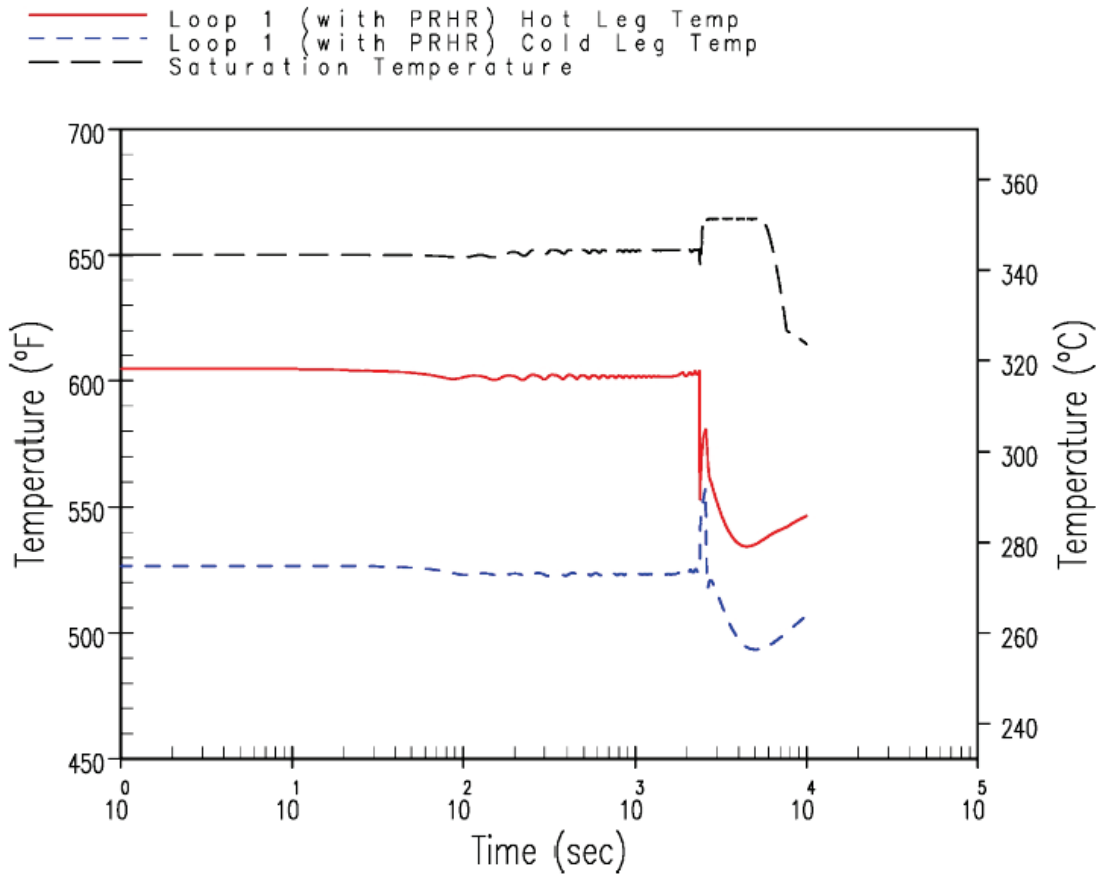


Figure 15.5.1-2
RCS Temperature Transient in Loop Containing the PRHR for Inadvertent Operation of the Emergency Core Cooling System Due to a Spurious Opening of the Core Makeup Tank Discharge Valves

Revised UFSAR Figure 15.5.1-2:



Updated UFSAR Figure 15.5.1-2
RCS Temperature Transient in Loop Containing the PRHR
for Inadvertent Operation of the Emergency Core Cooling System
Due to a Spurious Opening of the Core Makeup Tank Discharge Valves

Delete current UFSAR Figure 15.5.1-3:

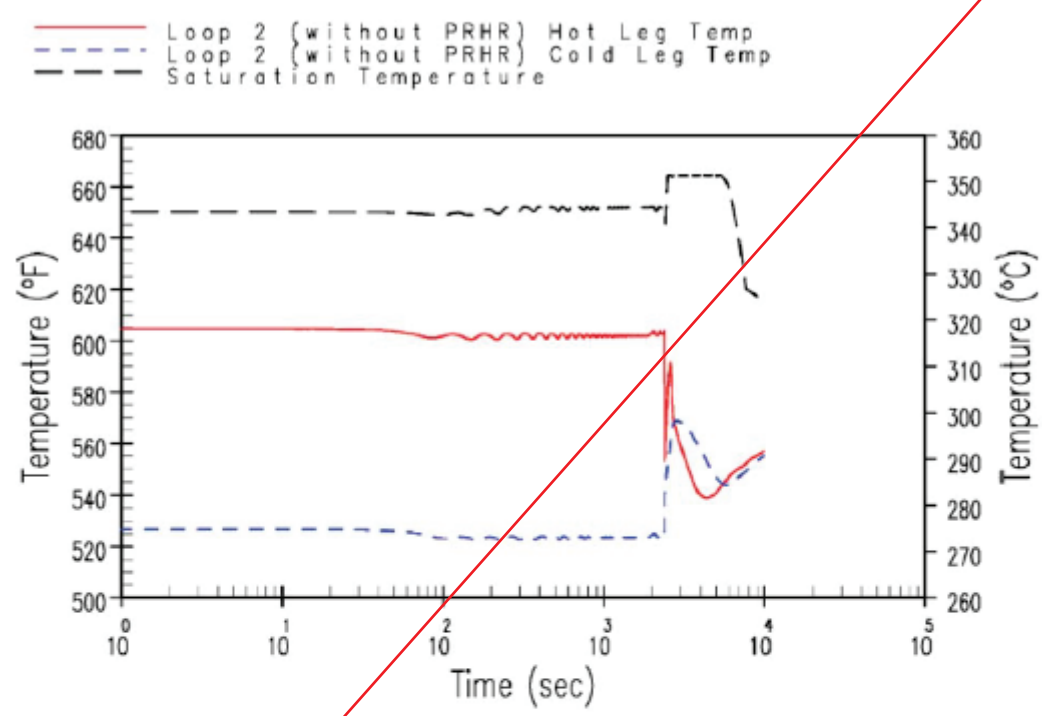
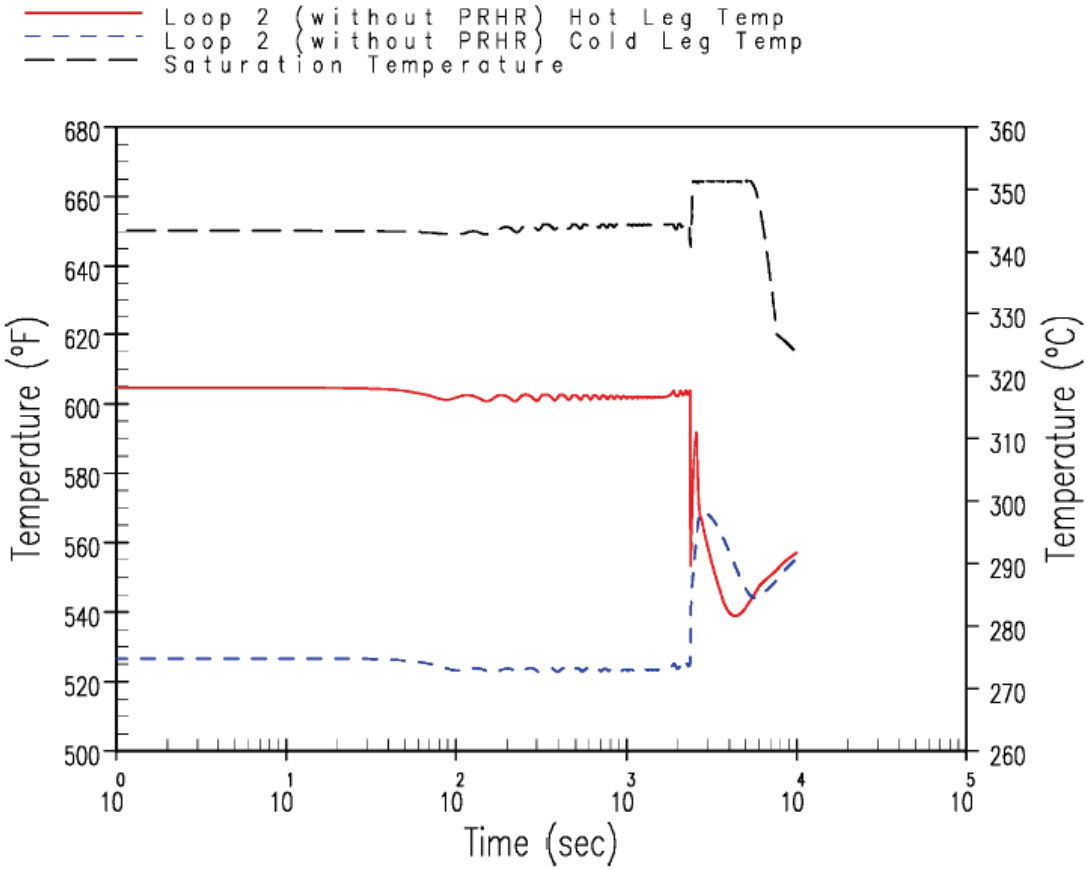


Figure 15.5.1-3
RCS Temperature Transient in Loop Not Containing the PRHR for Inadvertent Operation of the Emergency Core Cooling System Due to a Spurious Opening of the Core Makeup Tank Discharge Valves

Revised UFSAR Figure 15.5.1-3:



Updated UFSAR Figure 15.5.1-3
RCS Temperature Transient in Loop Not Containing the PRHR
for Inadvertent Operation of the Emergency Core Cooling System
Due to a Spurious Opening of the Core Makeup Tank Discharge Valves

Delete current UFSAR Figure 15.5.1-4:

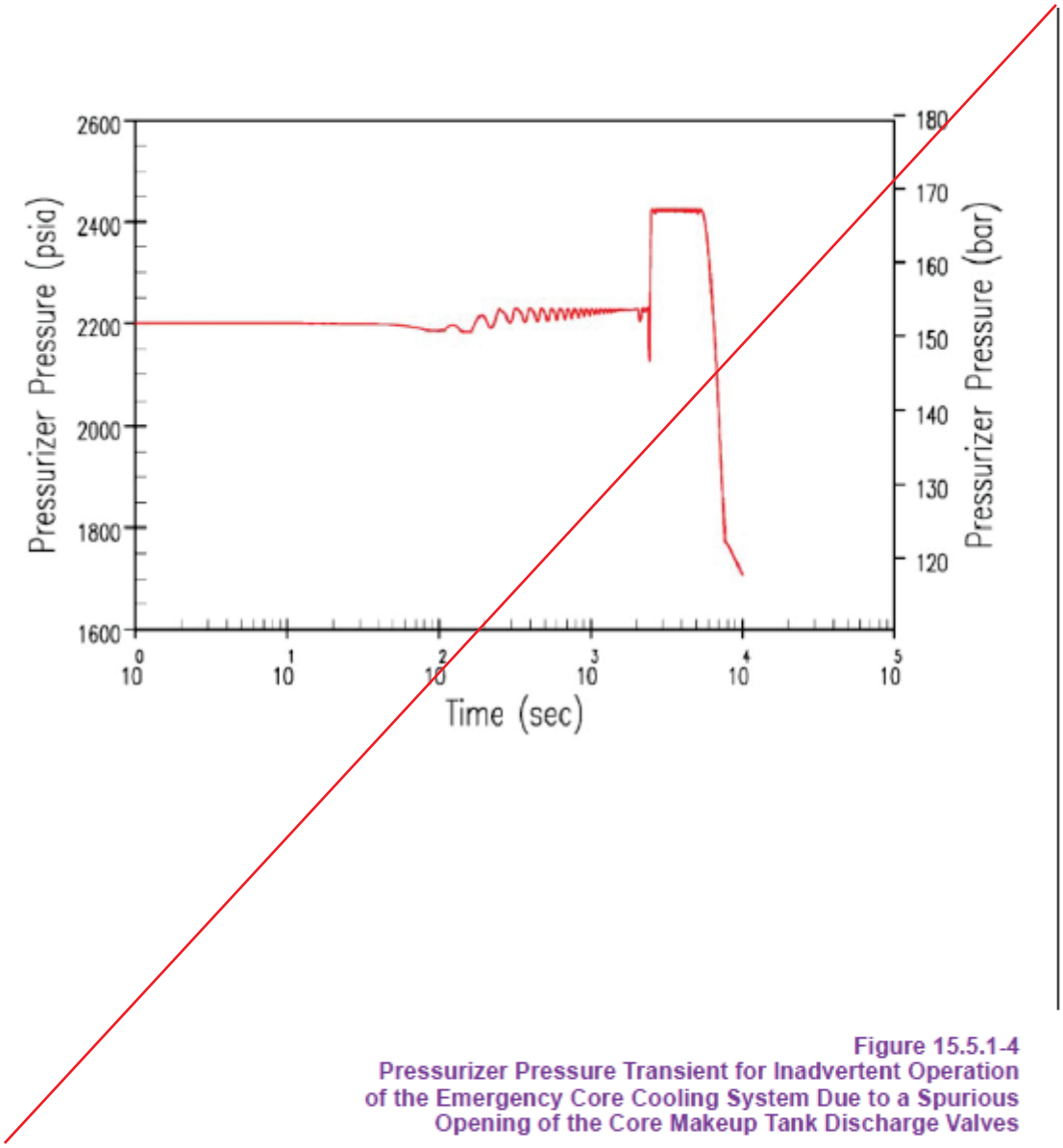
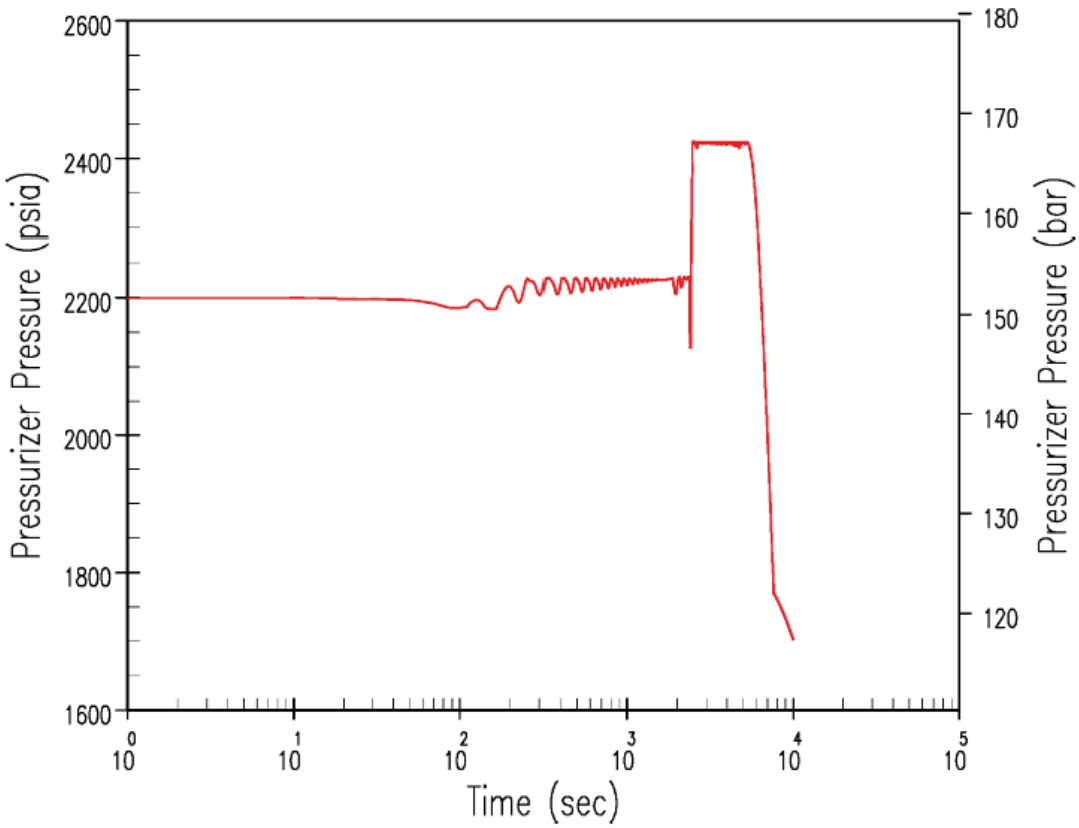


Figure 15.5.1-4
Pressurizer Pressure Transient for Inadvertent Operation of the Emergency Core Cooling System Due to a Spurious Opening of the Core Makeup Tank Discharge Valves

Revised UFSAR Figure 15.5.1-4:



Updated UFSAR Figure 15.5.1-4
Pressurizer Pressure Transient for Inadvertent Operation of the Emergency Core Cooling System Due to a Spurious Opening of the Core Makeup Tank Discharge Valves

Delete current UFSAR Figure 15.5.1-5:

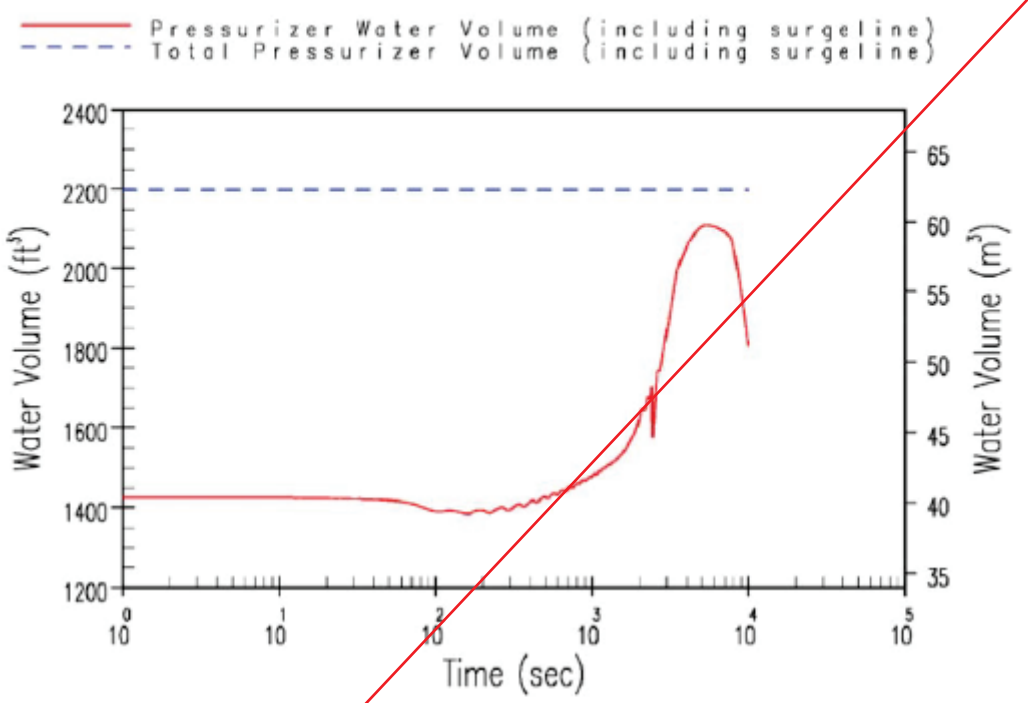
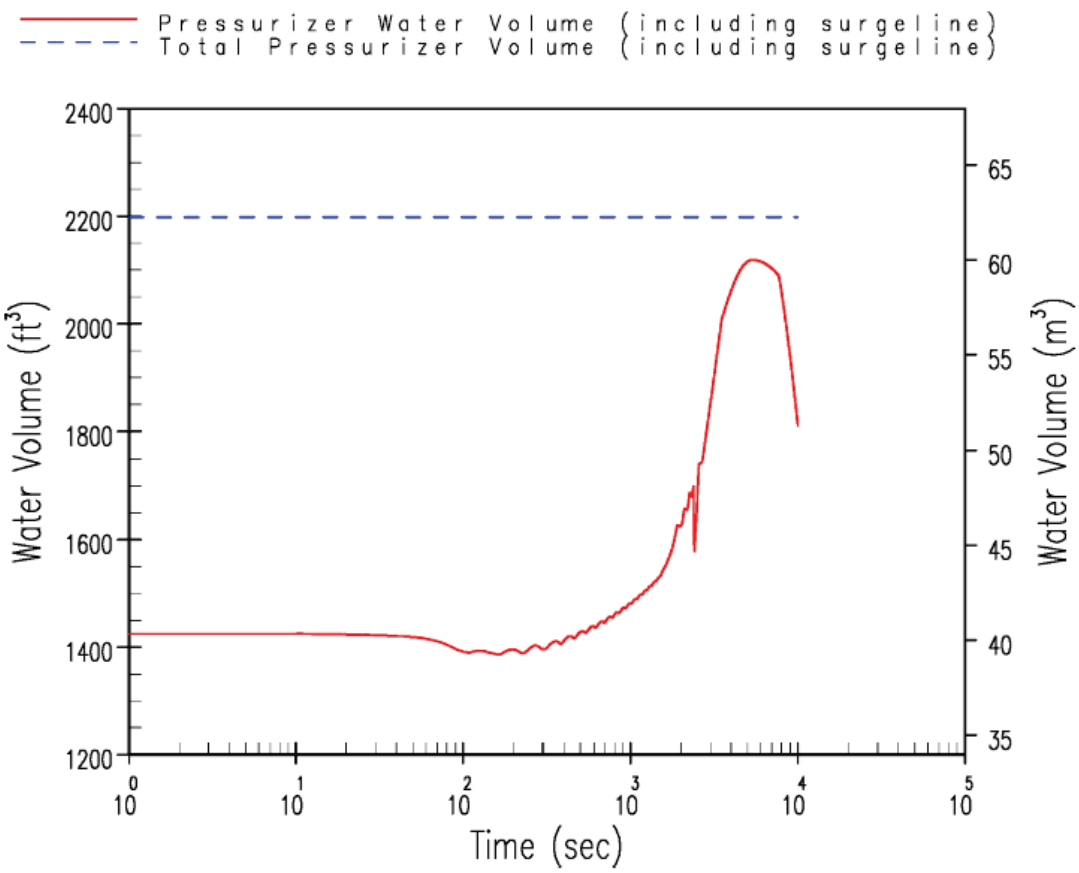


Figure 15.5.1-5
Pressurizer Water Volume Transient for Inadvertent Operation of the Emergency Core Cooling System Due to a Spurious Opening of the Core Makeup Tank Discharge Valves

Revised UFSAR Figure 15.5.1-5:



Updated UFSAR Figure 15.5.1-5
Pressurizer Water Volume Transient for Inadvertent Operation of the Emergency Core Cooling System Due to a Spurious Opening of the Core Makeup Tank Discharge Valves

Delete current UFSAR Figure 15.5.1-6:

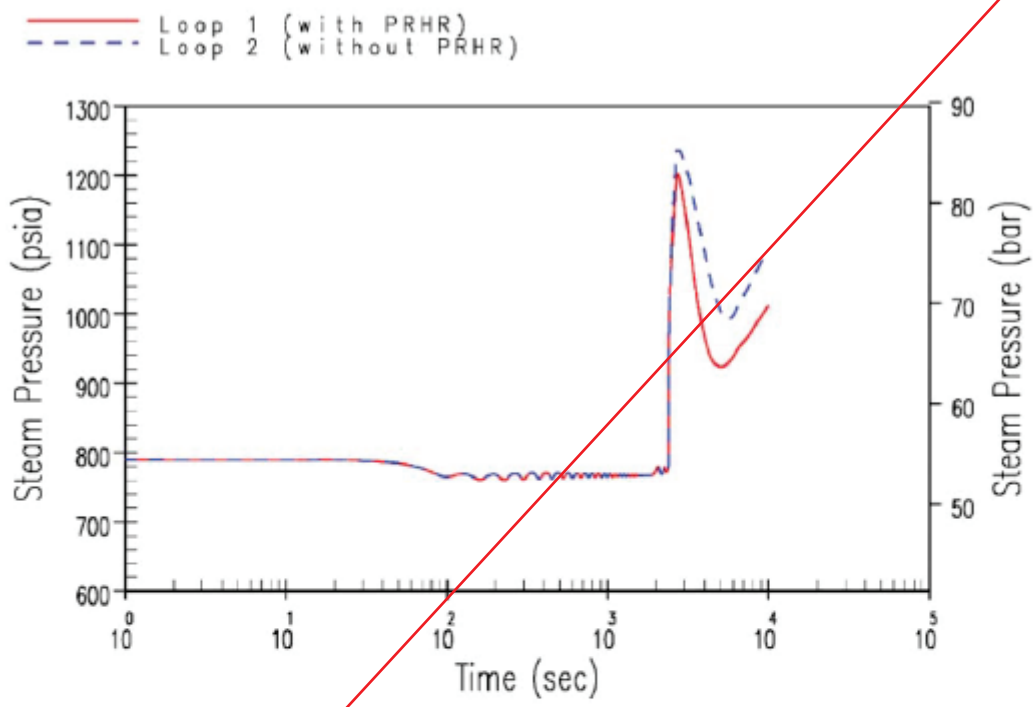
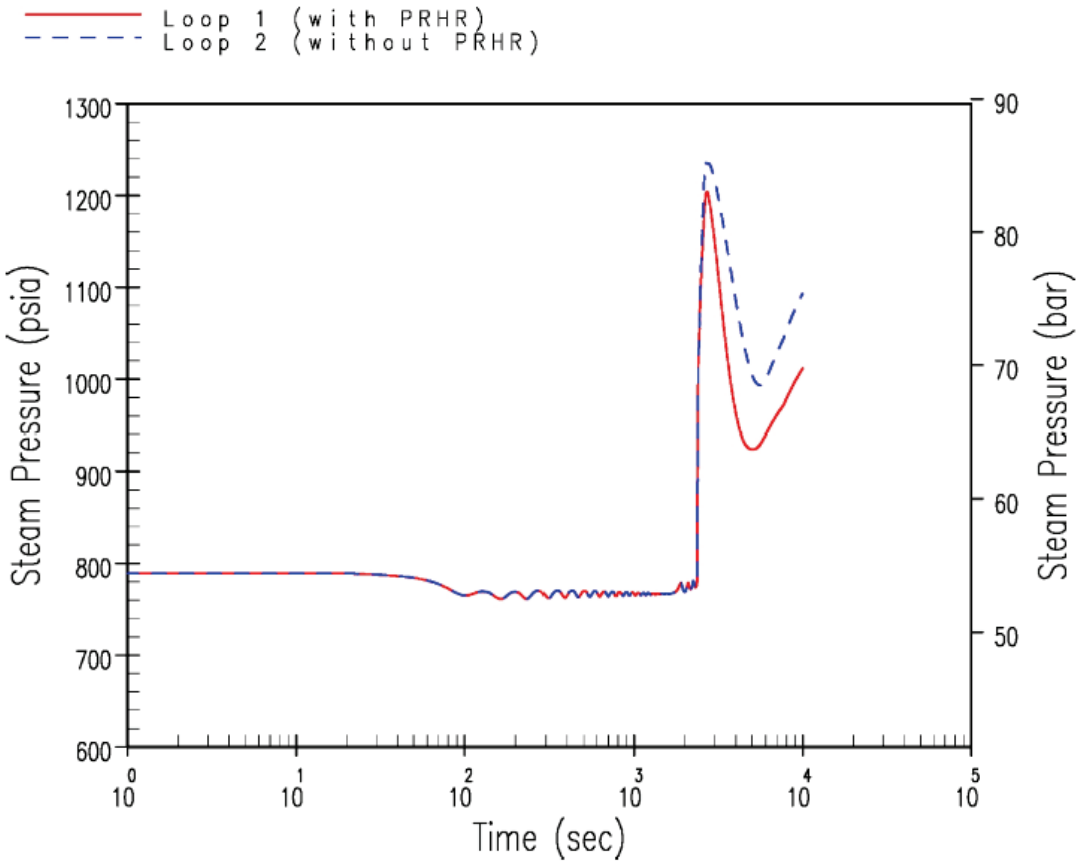


Figure 15.5.1-6
Steam Generator Pressure Transient for Inadvertent Operation of the Emergency Core Cooling System Due to a Spurious Opening of the Core Makeup Tank Discharge Valves

Revised UFSAR Figure 15.5.1-6:



Updated UFSAR Figure 15.5.1-6
Steam Generator Pressure Transient for Inadvertent Operation of the Emergency Core Cooling System Due to a Spurious Opening of the Core Makeup Tank Discharge Valves

Delete current UFSAR Figure 15.5.1-7:

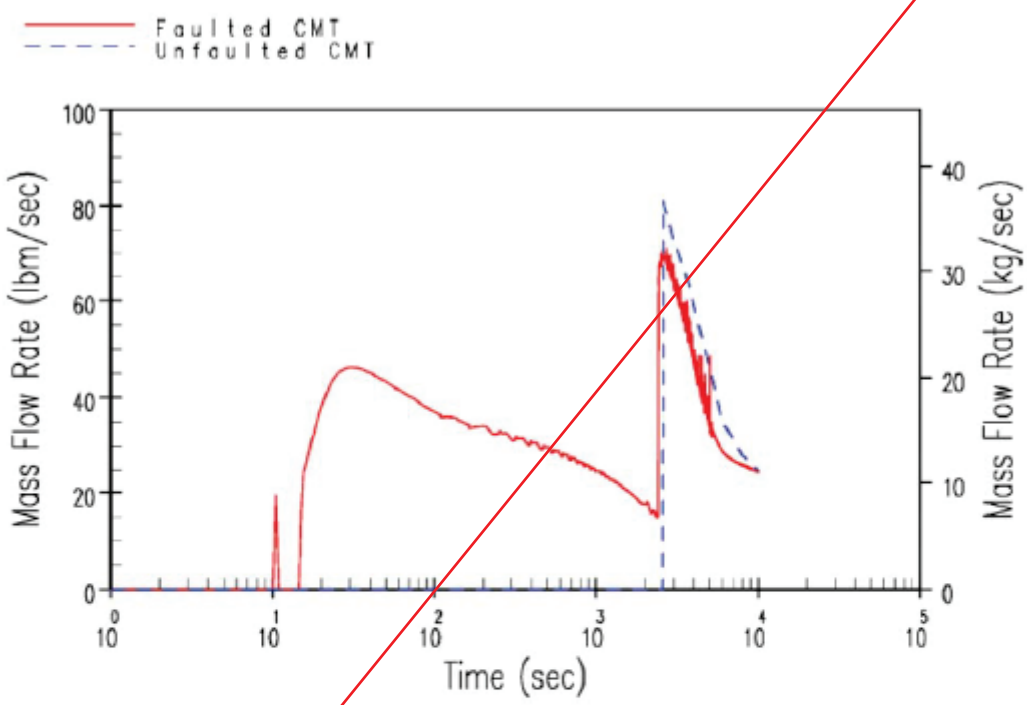
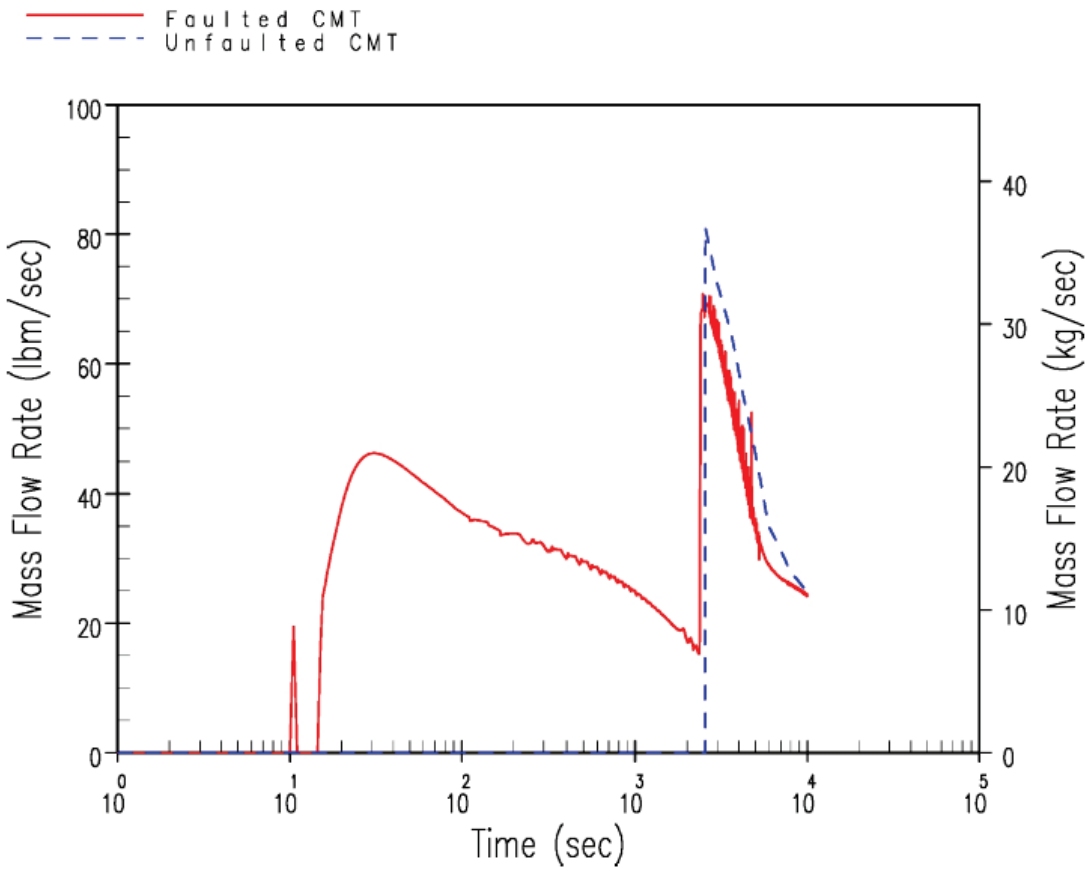


Figure 15.5.1-7
CMT Flow Rate Transient
for Inadvertent Operation of the Emergency Core Cooling System
Due to a Spurious Opening of the Core Makeup Tank Discharge Valves

Revised UFSAR Figure 15.5.1-7:



Updated UFSAR Figure 15.5.1-7
CMT Flow Rate Transient
for Inadvertent Operation of the Emergency Core Cooling System
Due to a Spurious Opening of the Core Makeup Tank Discharge Valves

Delete current UFSAR Figure 15.5.1-8:

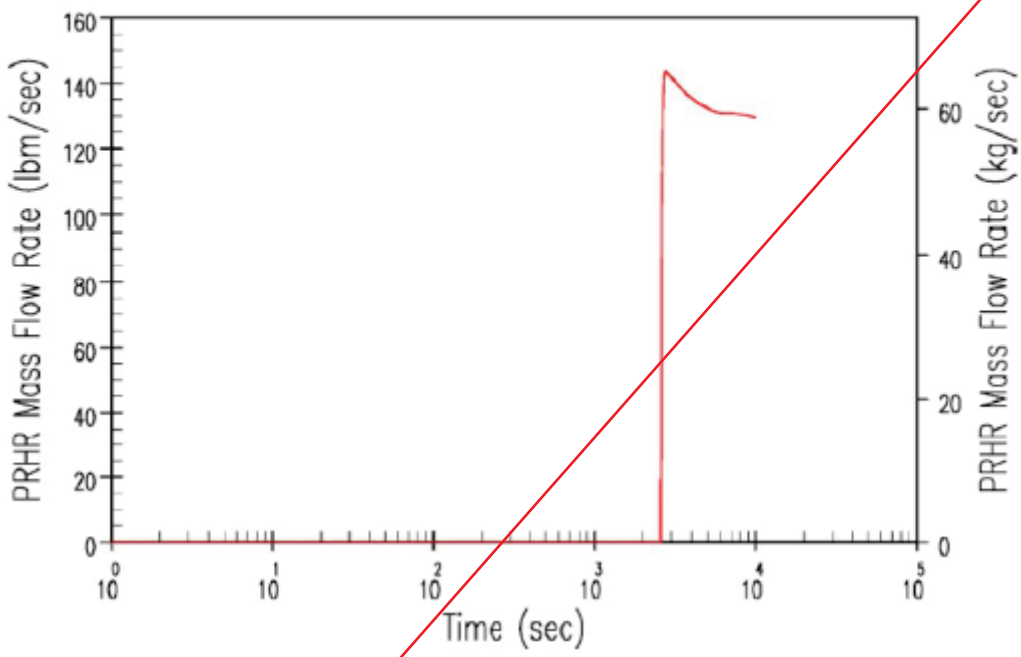
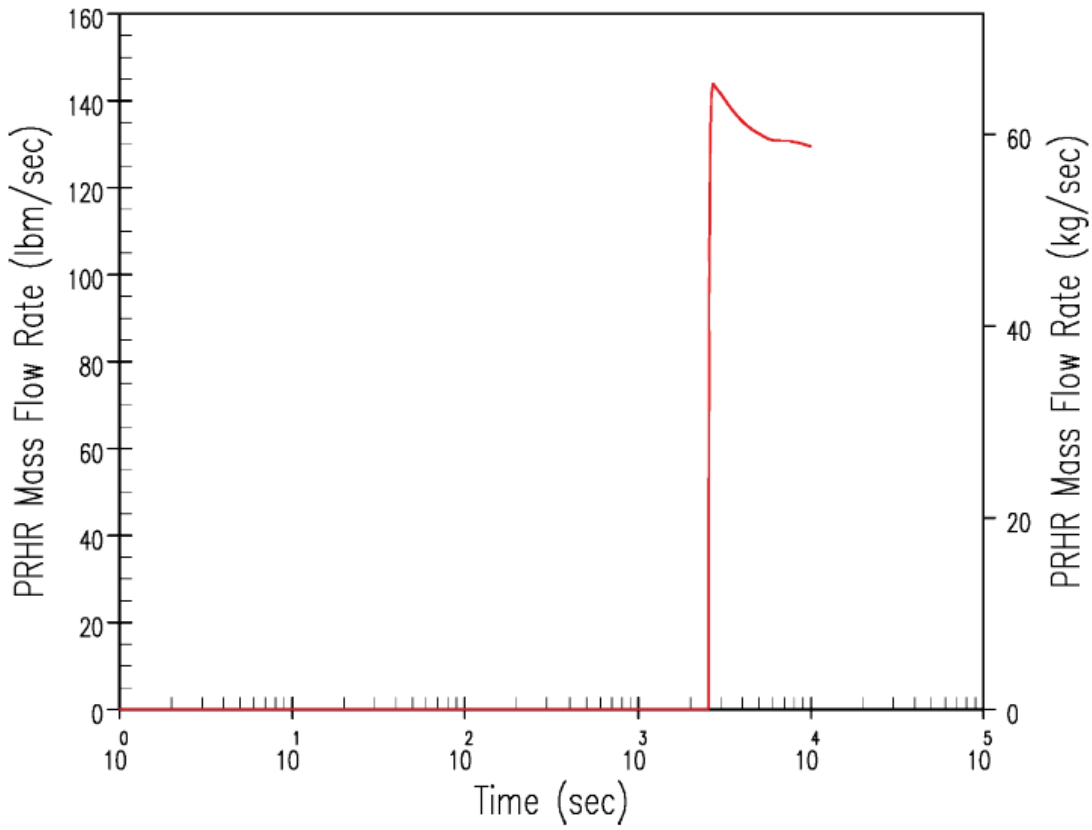


Figure 15.5.1-8
PRHR Flow Rate Transient
for Inadvertent Operation of the Emergency Core Cooling System
Due to a Spurious Opening of the Core Makeup Tank Discharge Valves

Revised UFSAR Figure 15.5.1-8:



Updated UFSAR Figure 15.5.1-8
PRHR Flow Rate Transient
for Inadvertent Operation of the Emergency Core Cooling System
Due to a Spurious Opening of the Core Makeup Tank Discharge Valves

Attachment 2
to the Enclosure of NL-25-0455

Revised Technical Specification Pages

3.5 PASSIVE CORE COOLING SYSTEM (PXS)

3.5.2 Core Makeup Tanks (CMTs) – Operating

LCO 3.5.2 Both CMTs shall be OPERABLE.

APPLICABILITY: MODES 1, 2, and 3.
MODE 4 with the Reactor Coolant System (RCS) not being cooled by the Normal Residual Heat Removal System (RNS).

ACTIONS

CONDITION	REQUIRED ACTION	COMPLETION TIME
A. One CMT inoperable due to one CMT outlet isolation valve inoperable.	A.1 Restore outlet isolation valve to OPERABLE status.	72 hours
B. One CMT inoperable due to water temperature or boron concentration not within limits.	B.1 Restore water temperature and boron concentration to within limits.	72 hours
C. Two CMTs inoperable due to water temperature or boron concentration not within limits.	C.1 Restore water temperature and boron concentration to within limits for one CMT.	1 hour from discovery of LCO 3.5.2 Condition C entry concurrent with LCO 3.5.1 Condition B entry AND 8 hours

ACTIONS (continued)

CONDITION	REQUIRED ACTION	COMPLETION TIME
D. One CMT inlet line with noncondensible gas volume not within limit.	D.1 Restore CMT inlet line noncondensible gas volume to within limit.	24 hours
E. One CMT inoperable for reasons other than Condition A, B, or D.	E.1 Restore CMT to OPERABLE status.	1 hour from discovery of LCO 3.5.2 Condition E entry concurrent with LCO 3.5.1 Condition B entry AND 8 hours
F. Required Action and associated Completion Time of Condition A, B, C, D, or E not met. OR Two CMTs inoperable for reasons other than Condition C.	F.1 Be in MODE 3. AND F.2 Be in MODE 5.	6 hours 36 hours

SURVEILLANCE REQUIREMENTS

SURVEILLANCE		FREQUENCY
SR 3.5.2.1	Verify the temperature of the borated water in each CMT is < 120°F.	24 hours
SR 3.5.2.2	Verify each CMT inlet isolation valve is fully open.	12 hours
SR 3.5.2.3	Verify the volume of noncondensible gases in each CMT inlet line has not caused the high-point water level to drop below the sensor.	24 hours

SURVEILLANCE REQUIREMENTS (continued)

SURVEILLANCE		FREQUENCY
SR 3.5.2.4	---------- **- NOTE -** One CMT boron concentration may be < 3400 ppm provided it is ≥ 3100 ppm and the average of OPERABLE CMT boron concentrations is ≥ 3400 ppm. ---------- Verify the boron concentration in each CMT is ≥ 3400 ppm and ≤ 4500 ppm.	31 days
SR 3.5.2.5	Verify each CMT outlet isolation valve strokes open.	In accordance with the Inservice Testing Program
SR 3.5.2.6	Verify each CMT outlet isolation valve actuates to the open position on an actual or simulated actuation signal.	24 months
SR 3.5.2.7	Verify system flow performance of each CMT in accordance with the System Level OPERABILITY Testing Program.	10 years

Attachment 3
to the Enclosure of NL-25-0455

Associated Technical Specification Bases Changes (For information only)

B 3.5 PASSIVE CORE COOLING SYSTEM (PXS)

B 3.5.2 Core Makeup Tanks (CMTs) – Operating

BASES

BACKGROUND

Two redundant CMTs provide sufficient borated water to assure Reactor Coolant System (RCS) reactivity and inventory control for all design basis accidents (DBAs), including both loss of coolant accident (LOCA) events and non-LOCA events (Ref. 1). Flow from the core makeup tank increases if the reactor coolant pumps (RCPs) have coasted down, therefore, the RCP breaker OPERABILITY is addressed in LCO 3.5.9, Reactor Coolant Pump Breakers.

The CMTs are cylindrical tanks with hemispherical upper and lower heads. They are made of carbon steel and clad on the internal surfaces with stainless steel. They are located in containment at an elevation slightly above the reactor coolant loops. Each tank is full of borated water at > 3400 ppm. During normal operation, the CMTs are maintained at RCS pressure through a normally open pressure balance line from the cold leg.

The outlet line from each CMT is connected to one of two direct vessel injection lines, which provides an injection path for the water supplied by the CMT. The outlet line from each CMT is isolated by parallel, normally closed, fail open valves. Upon receipt of a safeguards actuation signal, these four valves open to align the CMTs to the RCS.

The CMTs will inject to the RCS as inventory is lost and steam or reactor coolant is supplied to the CMT to displace the water that is injected. Steam or reactor coolant is provided to the CMT through the cold leg balance line, depending upon the specific event that has occurred. The inlet line from the cold leg is sized for LOCA events, where the cold legs become voided and higher CMT injection flows are required.

The injection line from each CMT contains a flow tuning orifice that is used to provide a mechanism for the field adjustment of the injection line resistance. The orifice is used to establish the required flow rates for the associated plant conditions assumed in the CMT design. The CMT flow is based on providing injection for approximately 20 minutes after CMT actuation.

The CMT size and injection capability are selected to provide adequate RCS boration and safety injection for the limiting DBA. One CMT at ≥ 3100 ppm is adequate for this function during a small break LOCA where one CMT completely spills via the pipe break (Ref. 2). The Probabilistic Risk Assessment (PRA) (Ref. 3) shows that none of the CMTs are required for small LOCAs, assuming that at least one accumulator is available.

BASES

APPLICABLE SAFETY ANALYSES

The CMTs are assumed to be OPERABLE to provide emergency boration and core makeup when the Chemical and Volume Control System (CVS) is unavailable or insufficient, and to mitigate the consequences of any DBA which requires the safety injection of borated water (Ref. 2).

Following a non-LOCA event such as a steam line break, the RCS experiences a decrease in temperature and pressure due to an increase in energy removal by the secondary system. The cooldown results in a reduction of the core SHUTDOWN MARGIN due to the negative moderator temperature coefficient, with a potential for return to power. The actuation of ~~the~~ both CMTs with average boron concentration ≥ 3400 ppm following this event provides injection of borated water to mitigate the reactivity transient and ensure the core remains shut down.

In the case of a steam generator tube rupture (SGTR), CMT injection provides borated water to compensate for RCS LEAKAGE.

In the case of an RCS leak of 10 gallons per minute, the CMTs can delay depressurization for at least 10 hours, providing makeup to the RCS and remain able to provide the borated water to compensate for RCS shrinkage and to assure the RCS boration for a safe shutdown.

In the case of a LOCA, the CMTs provide a relatively large makeup flow rate for approximately 20 minutes, in conjunction with the accumulators to provide the initial core cooling.

In the case of a small break LOCA, the borated water volume of one CMT is adequate for RCS safety injection, where one CMT completely spills via the pipe break.

CMTs satisfy Criterion 3 of 10 CFR 50.36(c)(2)(ii).

LCO

This LCO establishes the minimum conditions necessary to ensure that CMT parameters meet the initial conditions assumed in the safety analyses. OPERABILITY is not expected to be challenged due to small gas accumulations in the high point, and rapid gas accumulations are not expected during plant operation. However, a relatively small gas volume was incorporated into the design for alerting operators to provide sufficient time to initiate venting operations before the gas volume would be expected to increase to a sufficient volume that might potentially challenge the OPERABILITY of natural circulation flow. Therefore, noncondensible gas accumulation in the inlet line high point that causes the water level to drop below the sensor will require operator action to investigate the cause of the gas accumulation and to vent the associated high point(s).

BASES

LCO (continued)

Each CMT represents 100% of the total injected CMT borated water assumed in LOCA analysis. If the injection line from a single CMT to the vessel breaks, no single active failure on the other CMT will prevent the injection of borated water into the vessel. Thus the assumptions of the LOCA analysis with regard to the contribution of a single CMT will be satisfied with the CMT boron concentration ≥ 3100 ppm.

For non-LOCA analysis, two CMTs are assumed with average boron concentration ≥ 3400 ppm. Note that for non-LOCA analysis, the accident cannot disable a CMT.

APPLICABILITY

In MODES 1, 2, and 3, and MODE 4 when the RCS is not being cooled by the Normal Residual Heat Removal System (RNS) the CMTs are required to be OPERABLE to provide borated water for RCS inventory makeup and reactivity control following a design basis event and subsequent cooldown.

The CMT requirements in MODE 5 with the RCS pressure boundary intact are specified in LCO 3.5.3, “Core Makeup Tanks (CMTs) – Shutdown, RCS Intact.”

The CMTs are not required to be OPERABLE while in MODE 5 with the RCS pressure boundary open or in MODE 6 because the RCS is depressurized and borated water can be supplied from the In-containment Refueling Water Storage Tank (IRWST), if needed.

In the unlikely event of a total loss of AC power sources, coupled with an inoperable Passive Residual Heat Removal Heat Exchanger (PRHR HX) (beyond DBA), the CMTs may be used in a feed and bleed sequence to remove heat from the RCS.

ACTIONS

A.1

With one outlet isolation valve inoperable on one CMT, action must be taken to restore the valve. In this Condition, the CMT is capable of performing its safety function, provided a single failure of the remaining parallel isolation valve does not occur. A Completion Time of 72 hours is acceptable for a two train emergency core cooling system (ECCS) which is capable of performing its safety function without a single failure.

B.1

If the water temperature or boron concentration of one CMT is not within limits, it must be returned to within limits within 72 hours. The deviations in these parameters are expected to be slight, considering the frequent

BASES

SURVEILLANCE REQUIREMENTS (continued)

This level is well above the CMT level corresponding to that assumed in the safety analysis for CMT injected volume. Verifying the noncondensible gas volume is within the noncondensible gas limit confirms ≥ 2487 cu. ft. of borated water volume in the CMT.

The 24 hour Frequency is based on the expected low rate of gas accumulation and the availability of control room indication.

SR 3.5.2.4

The minimum boron concentration is 3400 ppm to meet the CMT safety analysis minimum reactivity control requirements. The maximum boron concentration allowed within each CMT at any time is 4500 ppm to prevent overboration.

The SR is modified by a Note that allows one CMT to meet its Surveillance (i.e., be considered OPERABLE) with boron concentration < 3400 ppm. This allowance requires that the CMT is ≥ 3100 ppm and that the average of both CMT boron concentrations is ≥ 3400 ppm. Furthermore, the higher boron concentration CMT must be OPERABLE (i.e., meeting other SRs). With the higher boron concentration CMT compensating for the boron reduction of one CMT, minimum reactivity control for non-LOCA break accidents would continue to be provided by injection of both CMTs. For LOCA events, boron concentration of the unaffected CMT ≥ 3100 ppm satisfies the minimum reactivity control assumption for boron addition from the CMT.

Verification every 31 days that the boron concentration in each CMT is within the required limits ensures that the reactivity control from each CMT, assumed in the safety analysis, will be available as required. The 31 day Frequency is adequate to identify changes which could occur from mechanisms such as in-leakage considering the provisions for monitoring temperature of the inlet line and top of the CMT. Additionally, the top of the CMT has control room indication and an alarm on increased temperature, which would be indicative of in-leakage.

SR 3.5.2.5

Verification that the redundant outlet isolation valves are OPERABLE by stroking the valves open ensures that each CMT will function as designed when these valves are actuated. Prior to opening the outlet isolation valves, the inlet isolation valve should be closed temporarily. Closing the inlet isolation valve ensures that the CMT contents will not be diluted or heated by flow from the RCS. Upon completion of the test, the inlet isolation valves must be opened. The Surveillance Frequency references the inservice testing requirements.